

FIRE MANAGEMENT

An aerial photograph of a forest landscape, likely a pine forest, showing the effects of different fire management treatments. The landscape is divided into three distinct sections. The left section, labeled 'Thinning Only', shows a forest with a higher density of trees, many of which appear to be dead or dormant, with a brownish-yellow hue. A road or path is visible winding through this area. The middle section, labeled 'Thinning + Prescribed Fire', shows a forest with a lower density of trees, appearing greener and more uniform in color. The right section, labeled 'No Treatment', shows a forest with a very high density of trees, appearing dark green and somewhat overgrown. The background features rolling hills and mountains under a cloudy sky.

Thinning Only

Thinning + Prescribed Fire

No Treatment

Fire Management

- The problem
- Integrated Fire Management
 - Stand-level management → pyrosilviculture
 - Landscape-level management
 - Linear fuel treatments
 - Primary fuelbreak network
 - WUI and Secondary fuelbreak network
 - Area-based fuel treatments

The problem

The problem

Over the past 50+ years, Southern Europe has seen a marked increase in the frequency and intensity of extreme wildfire events. This escalation is the result of interacting changes in **demographics, land use, climate, and fire use.**

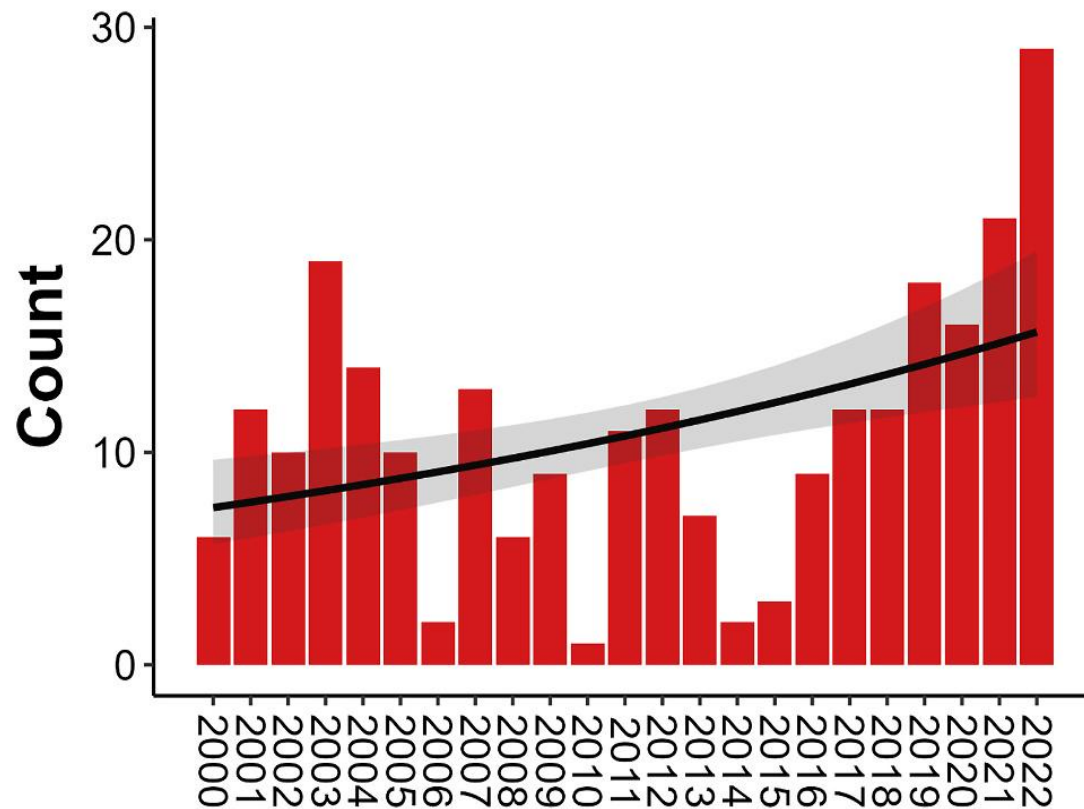


Fig. 6. Evolution of the number of contextually large fires (regardless of the return period) per year in Europe from 2000 to 2022. The trend is significant at p -value < 0.05 .

Demographic Changes

- **Rural depopulation:** Large-scale migration from rural to urban areas has led to land abandonment.
- **Loss of traditional land stewards:** As agricultural and pastoral populations declined, so did the active management of landscapes that once reduced fuel loads (e.g., grazing, clearing).
- **Urban sprawl into wildland areas:** Expansion of housing and infrastructure into wildland-urban interfaces (WUIs) has increased fire ignition risk and vulnerability.

Land Use Changes

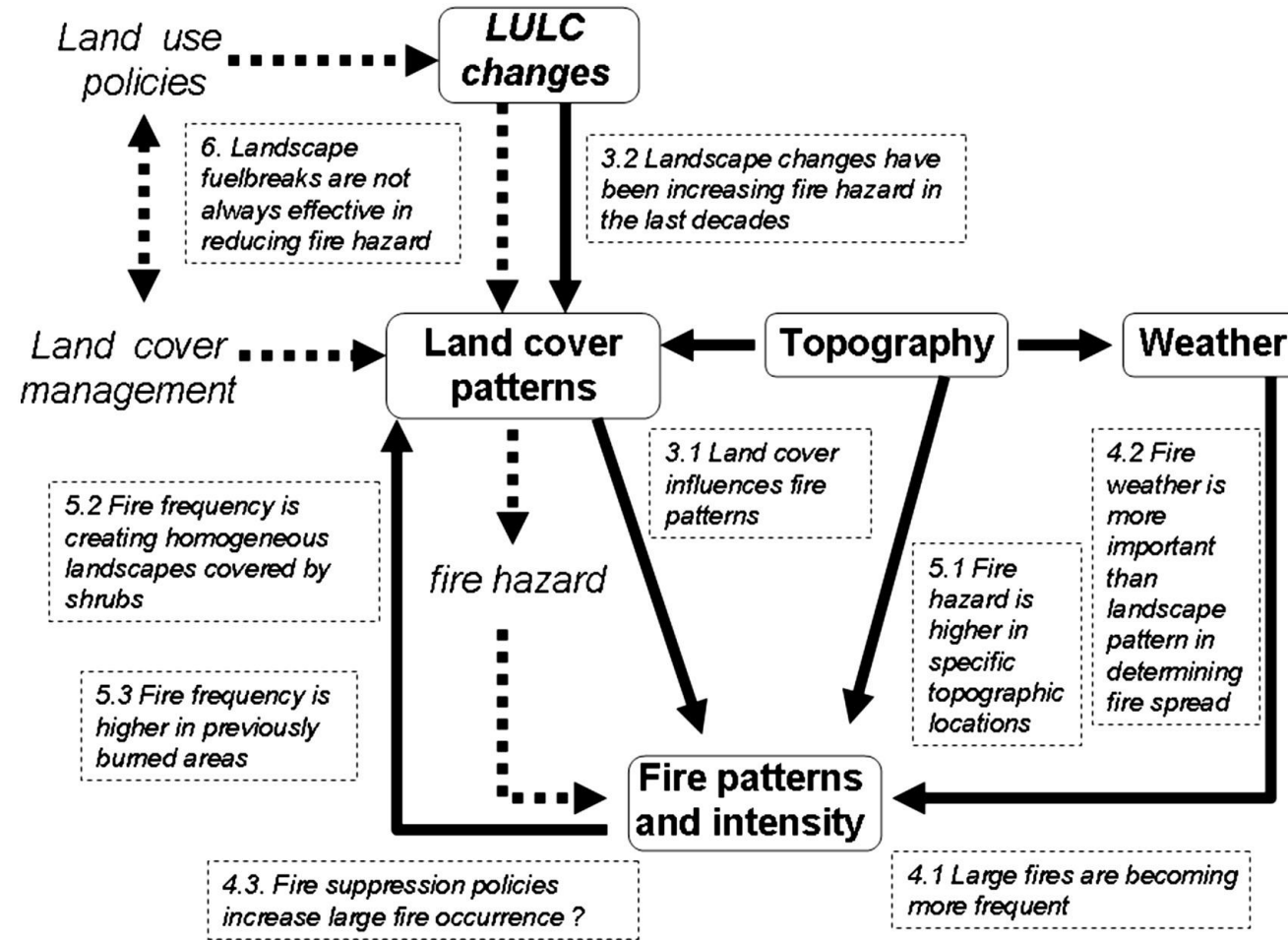
- **Agricultural abandonment:** Formerly cultivated lands have reverted to shrublands and young forests, increasing fuel continuity.
- **Afforestation and reforestation:** Often involving flammable species (e.g., Pinus, Eucalyptus), contributing to more fire-prone landscapes.
- **Homogenization of vegetation:** Reduced land-use diversity has led to more continuous and flammable vegetation patterns at the landscape scale.

Climate Change

- **Warmer temperatures:** Longer and hotter summers have extended fire seasons and dried fuels more thoroughly.
- **Reduced and erratic rainfall:** More frequent droughts and changing precipitation patterns increase vegetation stress and flammability.
- **Increased frequency of extreme weather events:** Heatwaves, dry lightning, and strong winds contribute to rapid fire spread and difficulty in containment.

Changes in Fire Use and Policy

- **Fire exclusion policies:** Decades of aggressive fire suppression have led to fuel build-up across landscapes.
- **Reduced traditional fire use:** Practices like pastoral burning and agricultural burning, once used to control vegetation, have been abandoned or restricted.
- **Delayed response to fuel accumulation:** Strategic prescribed burning and fuel reduction treatments are often underutilized due to regulatory, social, or resource constraints.



NOTE: the numbered boxes refer to paper sections,

- main drivers of fire spatial spread and intensity patterns across the landscape,
- - - - - management influences on these patterns..

Drivers	Landscape patterns	Fire hazard
Decreasing farming activities	Forests ↑ Shrubland ↑ Agricultural areas ↓	↑
Decreasing pastoral activities	Forests ↑ Shrubland ↑ Grassland and pasture ↓	↑
Population ageing and decline /emigration	Forests ↑ Shrubland ↑ Agricultural areas ↓	↑
Afforestation/reforestation	Forests ↑	↑
Decreased exploitation of timber and wood resources	Forests ↑ Shrubland ↑	↑
Increasing urban, tourist and industrial developments	Forests ↓ Shrubland ↓ Urban areas ↑	↓↑
Increasing farming activities	Forests ↓ Shrubland ↓ Agricultural areas	↓
Increasing pastoral activities	Forests ↓ Shrubland ↓ Grassland and pasture ↑	↓↑
Population increase/Immigration	Forests ↓ Shrubland ↓ Agricultural areas ↑ Urban areas ↑	↓↑

Table 1. Drivers of landscape change in European Mediterranean areas, their resulting landscape patterns, and implications for fire hazard. ↑ increase in area/fire hazard; ↓ decrease in area/fire hazard. Larger arrows in bold indicate the predominant trend for drivers with both increase and decrease in fire hazard.

Integrated fire management

Fire management

Fire management is the process of **planning, preventing** and **suppressing** fires to protect people, property, forests, and other ecosystems.



Fire exclusion paradigm → suppress all wildfires as quickly as possible to protect human life, property, and natural resources.

It assumes fire is inherently destructive and should be eliminated from landscapes.

Dominated much of the 20th century fire management policies, namely in the United States and Southern Europe.

It was the dominant paradigm in Portugal until 2017.



“A fire-free Portugal depends on everyone”

The fire paradox

- **Systematic fire suppression without adequate fuel reduction → fuel accumulation over time.**
- **Built-up fuels create conditions for → very large, high-severity wildfires.**
- **These fires usually occur when → suppression capacity is overwhelmed by severe weather.**



...or air-based wildfire suppression? 😊

The fire paradox

- **Fire regimes shift** → total burned area concentrates in **fewer, more extreme** fires, which are harder to fight.
- Also known as the **"firefighting trap"** → management system becomes **"victim of its own success"**.
- The main causes of the trap are:
 - land use changes leading to increased fire hazard and risk
 - persistence of reactive and shortsighted fire management policies



Drivers	Med basin	North America	Chile	South Africa	Australia
Agricultural land abandonment	H	L	L	L	L
Expansion of mismanaged tree plantations	H	L	H	M	L
Expansion of the wildland-urban interface	M	H	M	H	H
Invasive species	L	L	H	H	L
Hindered use of fire	M	H	L	H	H*
Lack of adequate post-fire management	M	M	M	L	M

*Mainly in eastern Australia, as there is a strong use of fire in Western Australia.

Figure 2. Drivers of the firefighting trap: estimated relative importance (coding: H=high; M=moderate; L=low) of major drivers the ‘firefighting trap’ across Mediterranean-type climate regions (MCRs), as evaluated using expert-knowledge (set of authors of this paper). Relative importance for each MCR inferred from potential share of the total area of the region affected by the driver and the resulting increases in fire hazard and exposure.

The fire-fighting trap

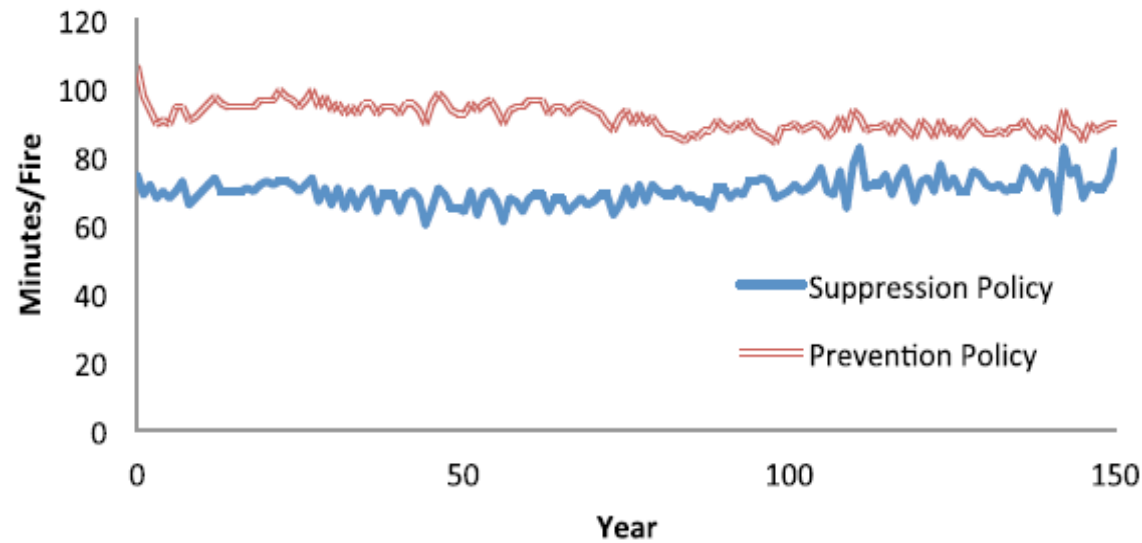


Fig. 3. **Fire Duration.** Fire durations (minutes/fire) are shorter under the Suppression Policy, which is the **intended effect** of the policy.

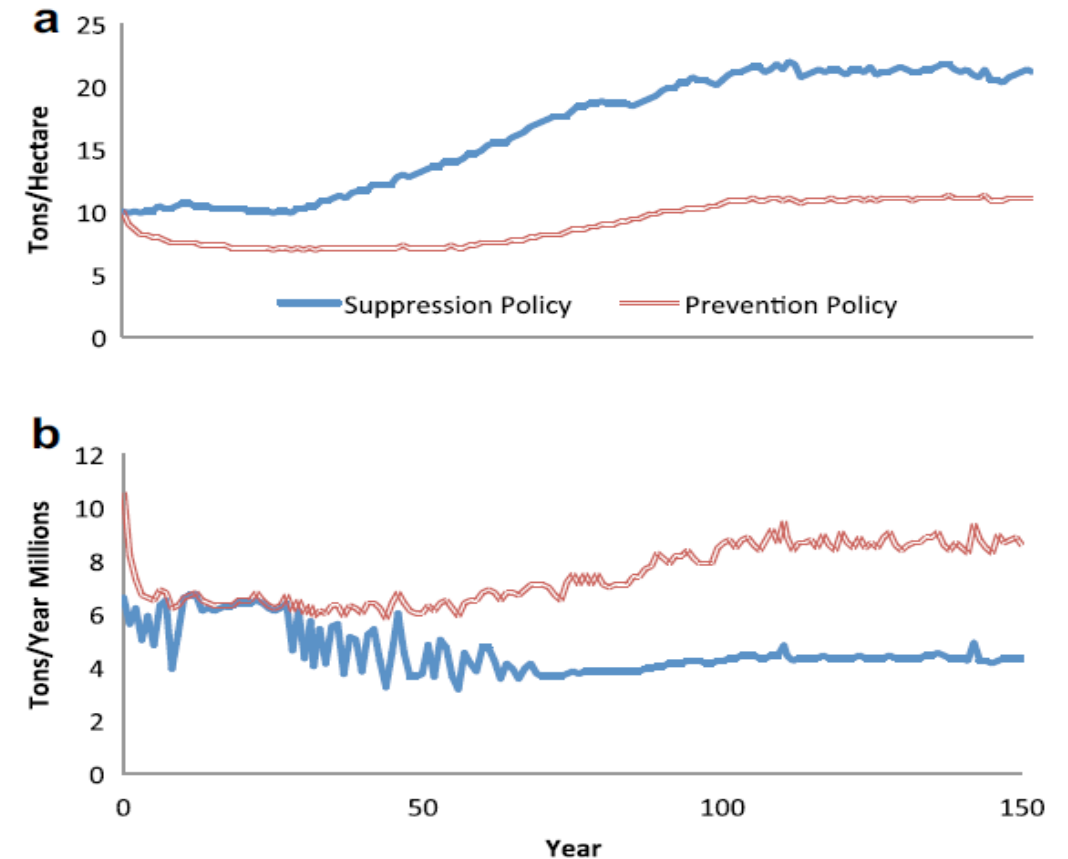


Fig. 4. a: **Fuel Load.** The Suppression Policy is associated with greater growth in fuel load (tons/hectare) compared to the Prevention Policy, which is the **unintended effect** of the policy.
b: **Preventative Removal.** With a finite budget, excessive suppression expenditure under the Suppression Policy crowds out preventative removal. **As the rate of preventative removal (tons of fuel per year) decreases, fuel load increases.**

The fire-fighting trap

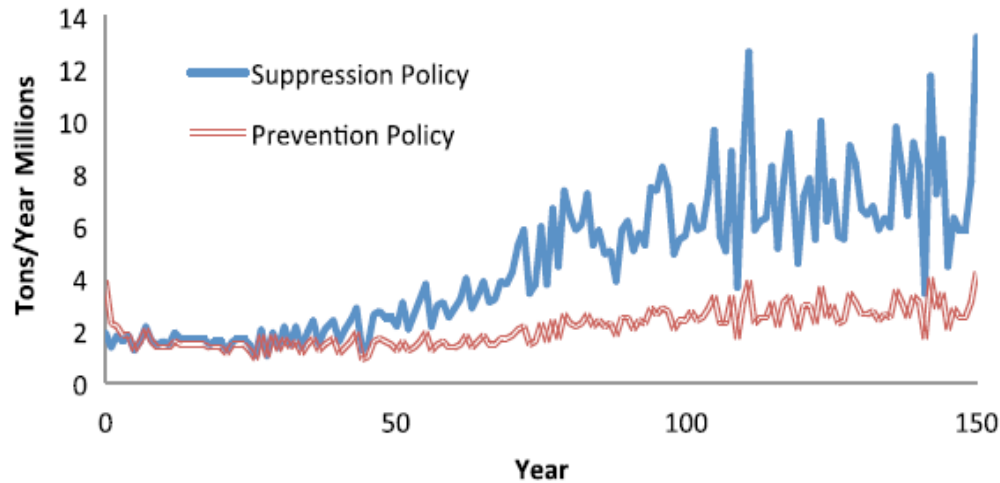


Fig. 5. **Forest Burning.** The rate of forest burning (tons of fuel per year) becomes larger and more volatile under the Suppression Policy.

NOTE: the high interannual variance of biomass burned under the Suppression policy occurs because fires are successfully extinguished in mild fire weather years. Biomass accumulates and burns very extensively during severe fire weather years.

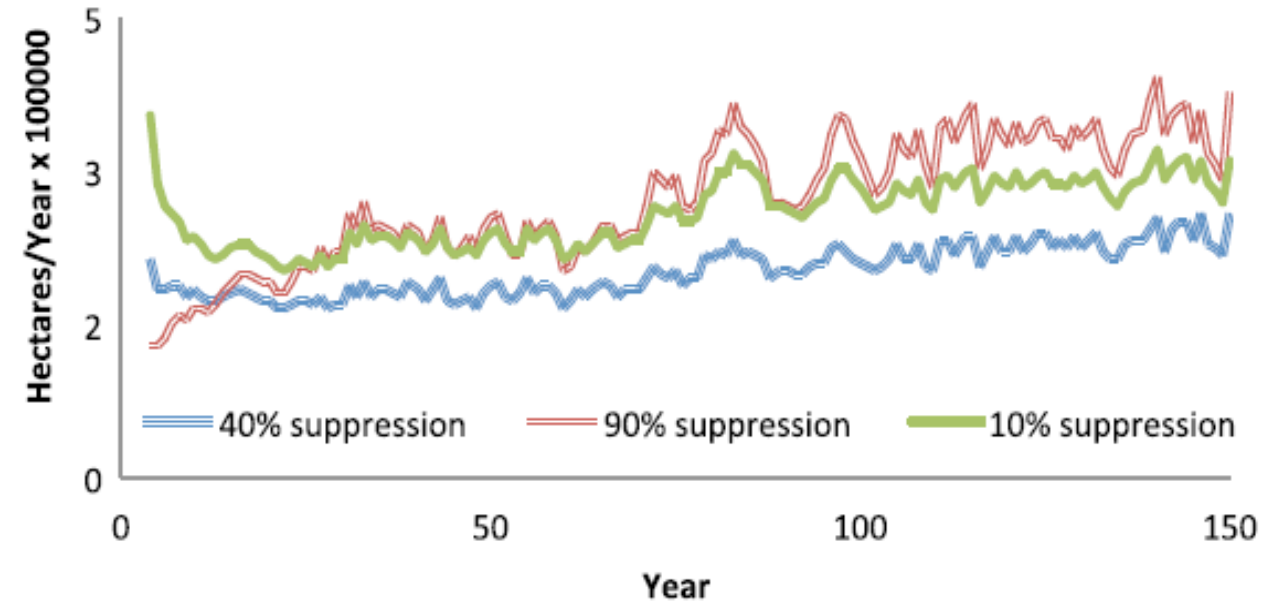


Fig. 7. **Burned Area per Year, 5-year moving average.** The appropriate balance of suppression and prevention efforts can greatly reduce yearly and total burned area.

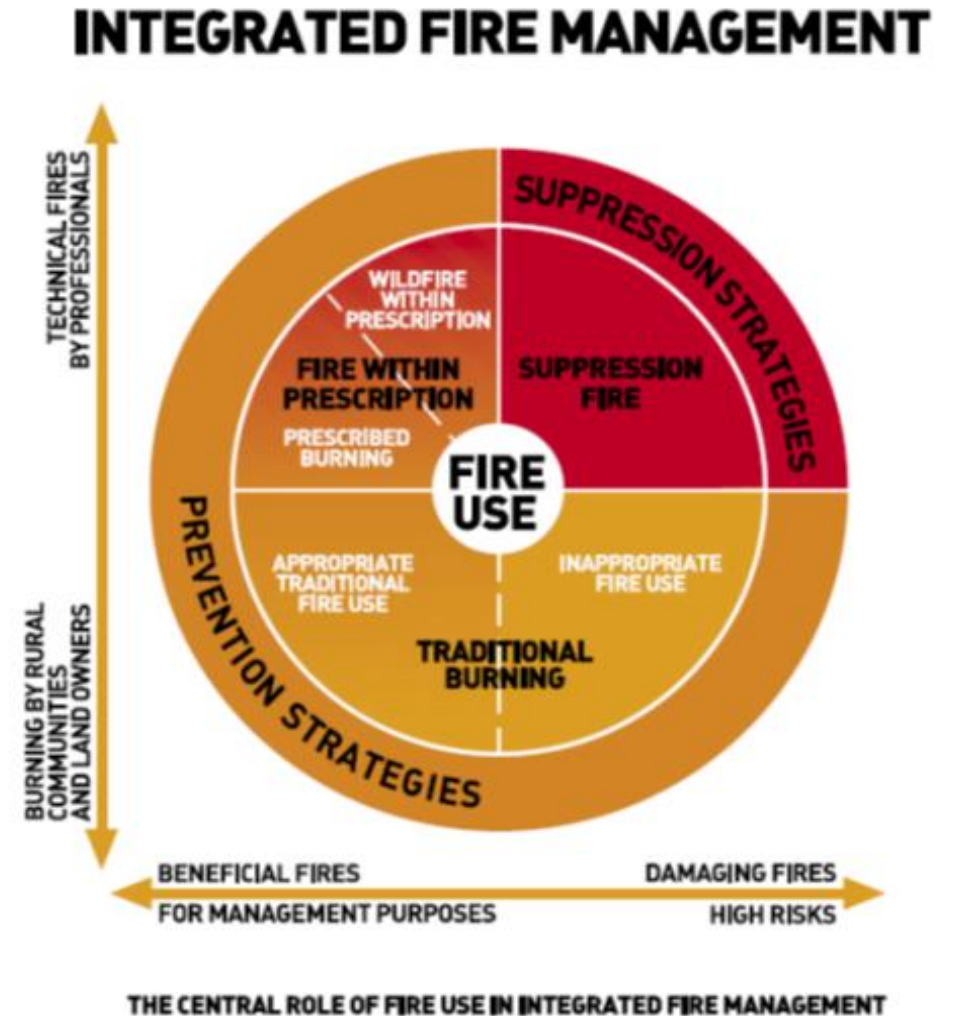
NOTE: the most effective policy is that with the more balanced budget between Prevention (60%) and Suppression (40%).

Integrated fire management

Emerged as an alternative to the fire exclusion paradigm and acknowledges the existence of “good fire” and “bad fire”.

It is a concept for planning and operational systems that include social, economical, cultural, and ecological evaluations with the objective of minimizing the damage and maximizing the benefits of fire.

Includes a combination of prevention and suppression strategies and techniques that integrate the use of technical fires and regulate traditional burning.



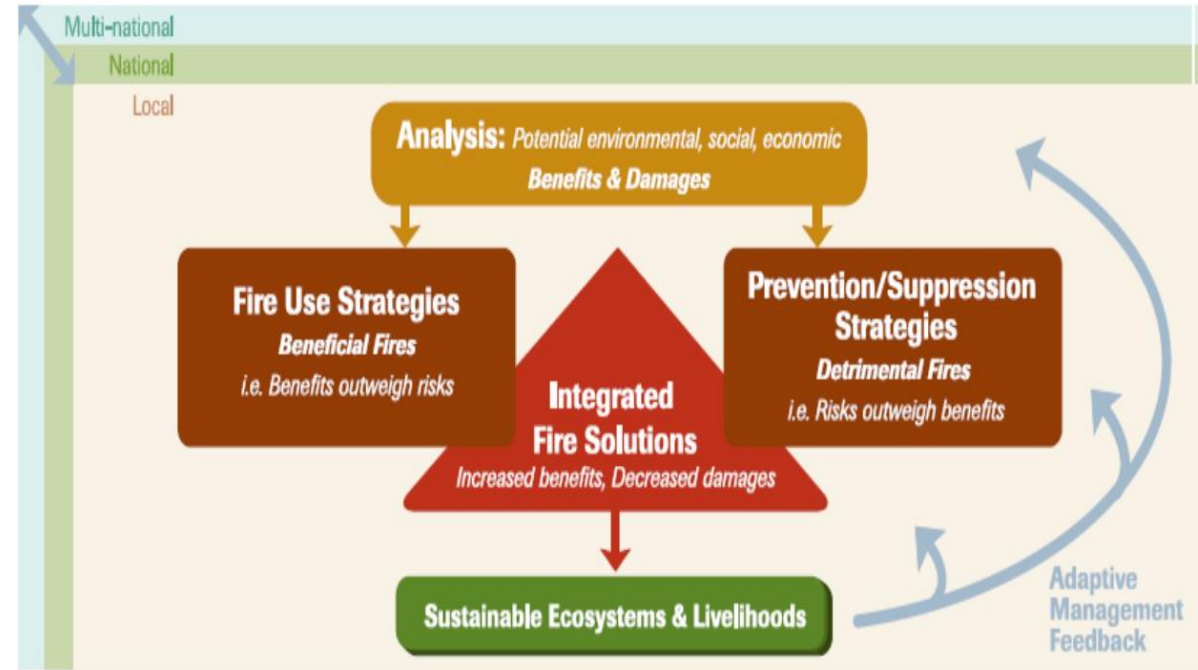
Integrated fire management

1. The **assessment and analysis of needs and issues**;
2. The identification of **fire management goals** and desired future condition;
3. The development and implementation of **appropriate fire policy**;
4. The design of fire **prevention/suppression strategies** appropriate to the ecological and social situation;



Integrated fire management

5. The implementation of **fire use strategies**;
6. The strengthening of **fire preparedness and response** at all levels;
7. The implementation of **recovery, restoration and ecosystem maintenance** actions; and
8. **Research** (e.g. fire ecology, fire behavior, social sciences).



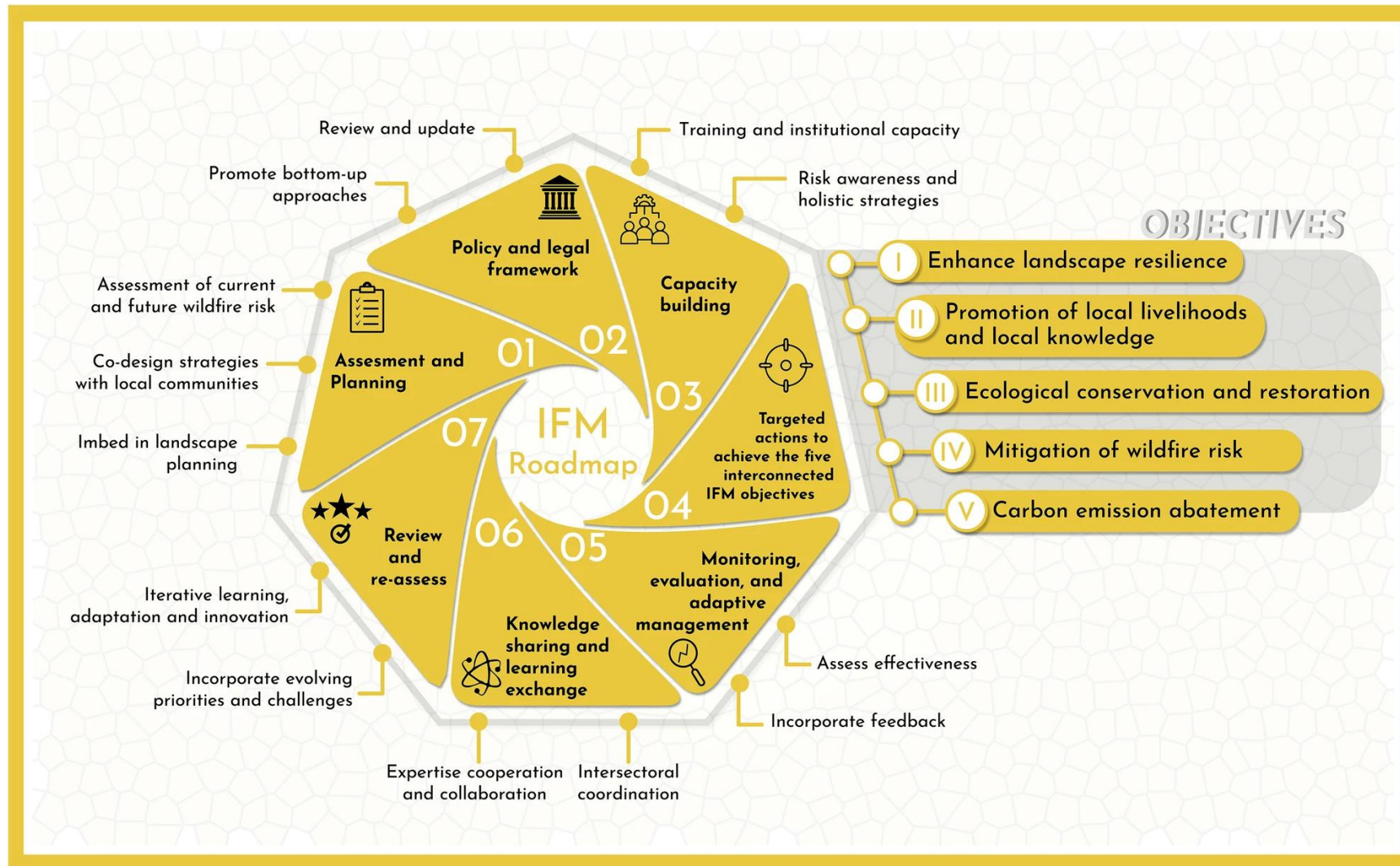
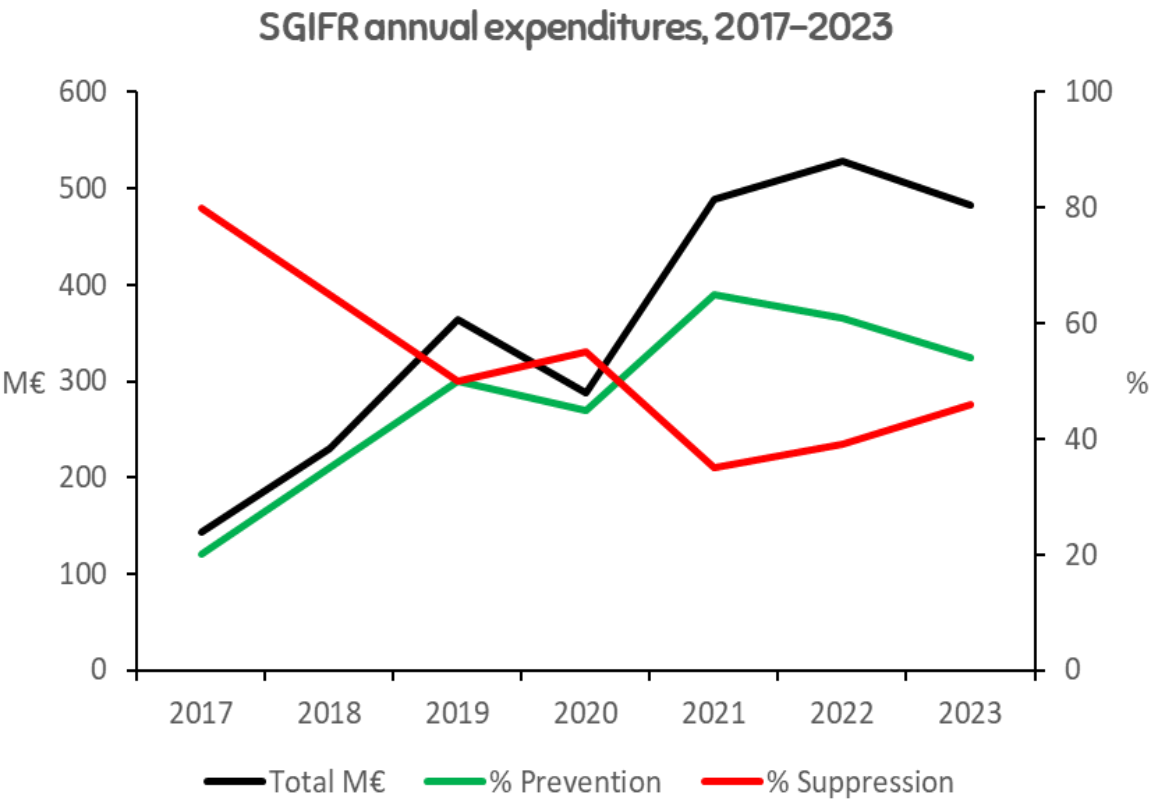


Fig. 2. Roadmap to developing IFM strategies aimed at adaptation and mitigation to altered fire regimes. This roadmaps involves seven core steps (see section 'Roadmap for Countries') and five main objectives (see section 'Setting IFM as an adaptation and mitigation strategy to altered fire regimes' for a description).

Integrated fire management



Annual fire management expenditures in Portugal have been evolving towards the appropriate balance between Prevention and Suppression recommended by the 2013 MIT study.

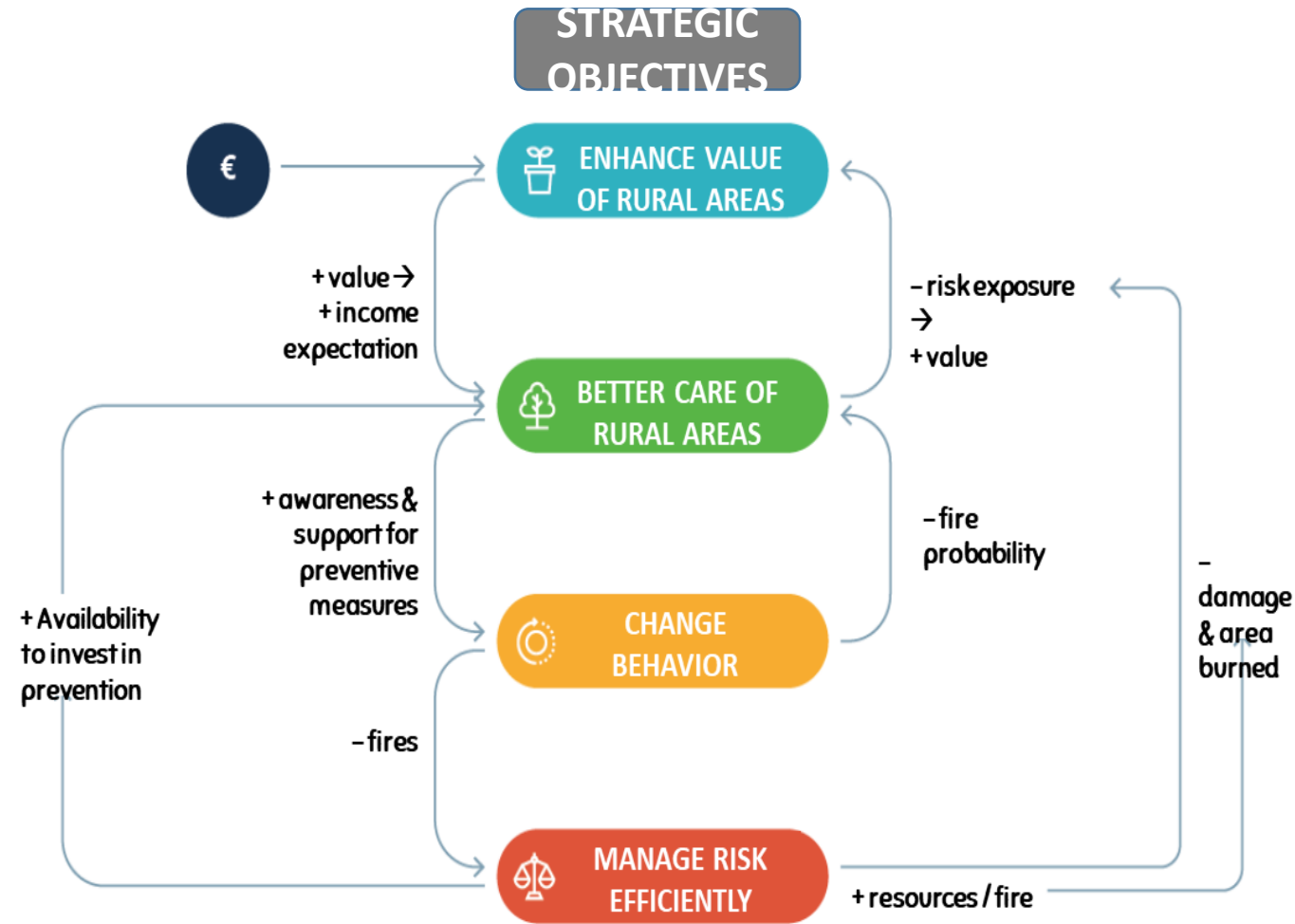
Portugal's National Plan for Integrated Rural Fire Management.

PNGIFR IFM approach to address rural fire management in Portugal:

A cascade of positive reinforcement cycles
→ efficient risk management → reduction
in area burned and fire damage.

Strategic objectives address:

- rural development
- stand- and landscape-level fuel management
- fire use behaviors
- fire risk management



Stand-level management (pyrosilviculture)

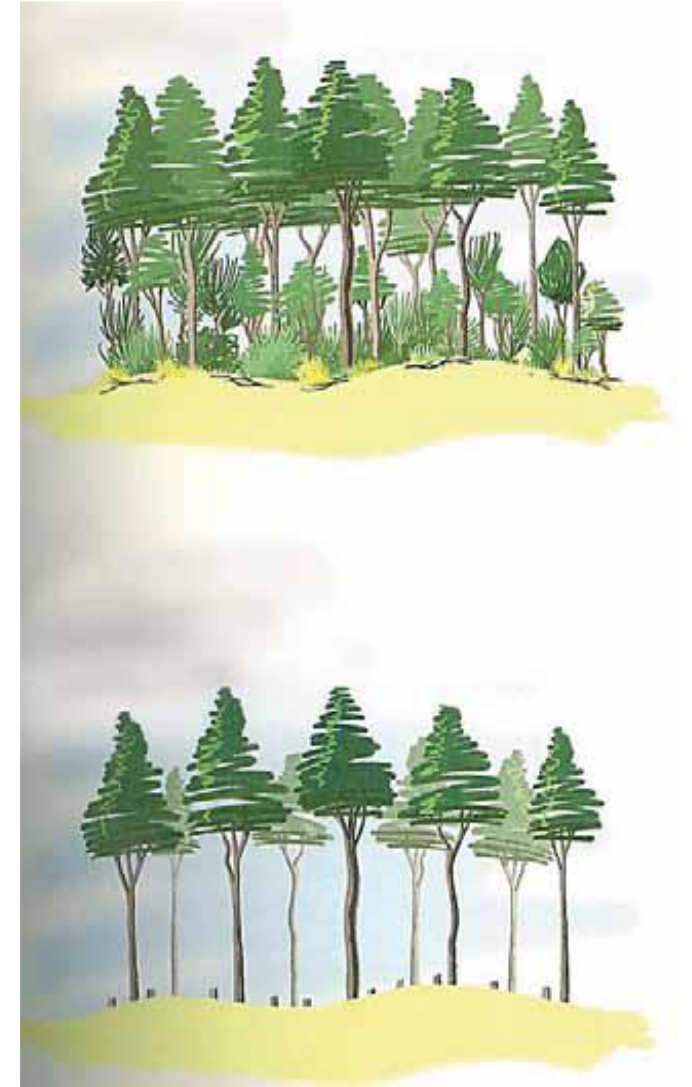
Preventive silviculture

The objective of preventive silviculture is to obtain a composition and structure of forest stands with the objective to:

- reduce the fire rate of spread and intensity;
- limit the tree damage caused by the fire passage.

These two objectives are closely interdependent. All measures aiming at reducing the fire intensity try to limit the damage caused to trees.

If the fire remains at the surface crowns will be less affected and chances of survival increase for a greater number of trees in the stand.



Principles of Fire-Resistant Forests

A fire-resistant forest makes crown fires unlikely and allows the forest to survive surface fire without significant tree mortality in the canopy.

We can lower fire risk and wildfire damage by removing or reducing fuels in strategic locations.

We can influence forest fuels so that fire becomes more manageable and less damaging.

Surface fuels and height to canopy are two factors that affect the transition from a surface fire to a canopy fire.



Principles of Fire-Resistant Forests

There are five principles of creating and maintaining fire-resistant forests:

- Reduce surface fuels
- Increase the height to the base of tree crowns
- Increase spacing between tree crowns
- Keep larger trees of more fire-resistant species
- Promote fire-resistant forests at the landscape level



Principles of Fire-Resistant Forests

By following these principles we can:

- Reduce the intensity of a fire, making it easier for firefighters to suppress.
- Increase the odds that larger proportions of a forest will survive a fire. Small trees, shrubs, and other understory vegetation may be injured or killed, but larger trees in the stand will only be scorched, and soil damage also will be reduced.
- Reduce extent of post-fire restoration activities needed, such as replanting.



Reduce surface fuels

Reducing surface fuels, such as slash and small shrubs, impairs potential flame length and fire intensity, making fires easier to control and less likely to reach into tree crowns.

Reducing surface fuels means removing significant accumulations of flammable organic material, but not eliminating all organic material down to mineral soil.



Increase distance to base of tree crowns

When tree crowns ignite (torching), the stage is set for a crown fire. Removing ladder fuels, including surface fuels, and pruning the larger trees raises the base of the forest canopy so that a longer flame is needed to ignite the crowns.

Pruning is particularly effective in young stands, when crowns may still be low to the ground.

Prescribed underburning can also increase the height of the lower canopy due to scorching and killing of lower branches.



Increase spacing between tree crowns

Thinning reduces crown density. When tree crowns are farther apart, it is harder for fire to spread from one crown to another, even when the wind is blowing.

Reducing the slash generated from thinning will diminish the potential for a high-intensity surface fire.

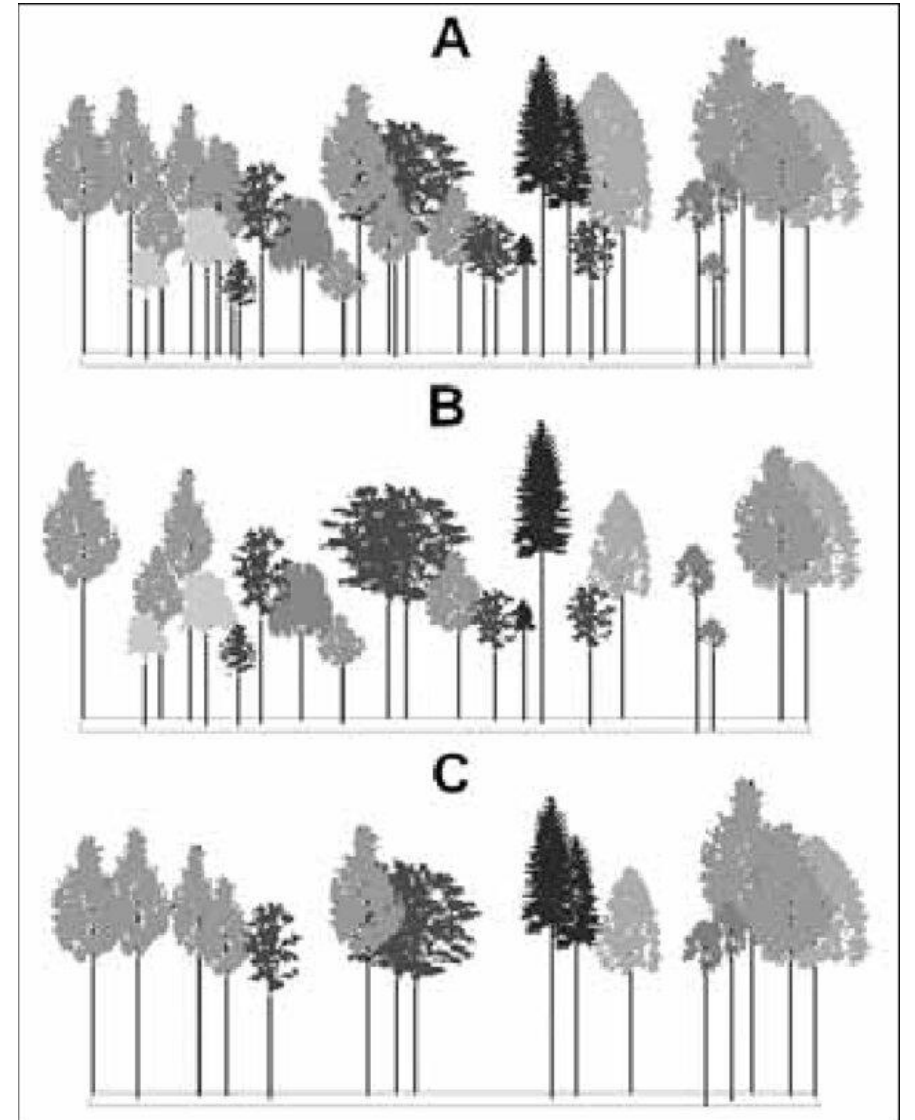


Keep large trees of more fire-resistant species

Fire kills trees by killing the cambium layer, scorching the foliage and killing the buds, and damaging and killing roots.

When thinning to improve fire resistance, leave larger trees with thicker bark that insulates the cambium. Although a fire may scorch the foliage above, the cambium is still protected.

Also, large trees tend to have higher crowns, so their foliage and buds are less likely to be damaged by heat from a surface fire.



Promote fire-resistant forests at the landscape level

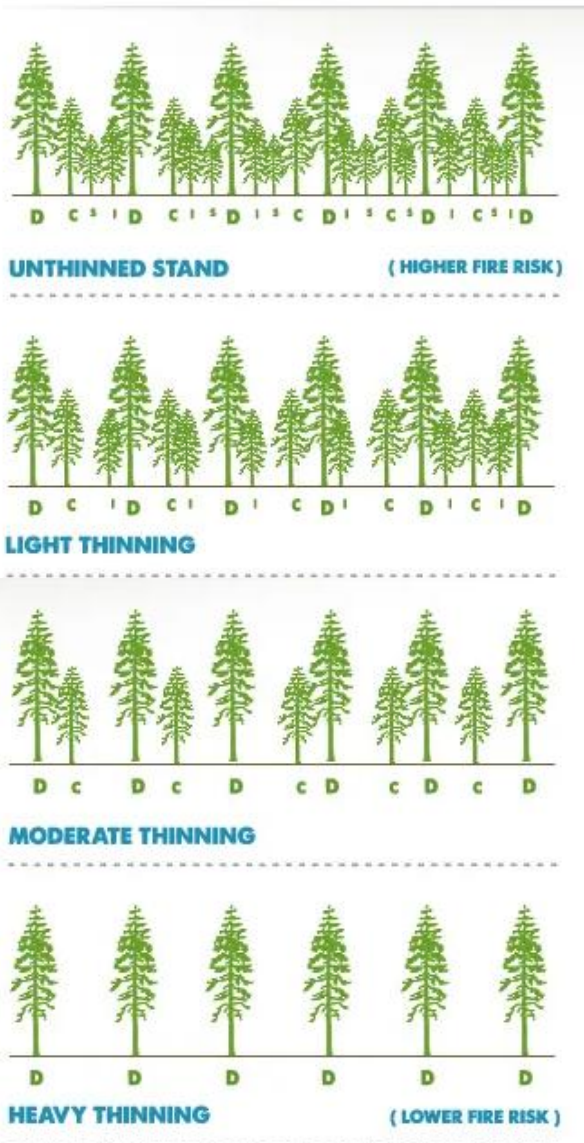
The larger an area treated, the more effective fuels treatments will be at moderating fire behavior. This includes creating gaps and openings to further reduce the potential for crown fire.

Treating in strategic locations can help break up both continuous horizontal and vertical layers of fuels.

For example, reducing fuels adjacent to natural features, such as meadows and rock outcroppings, and manmade features, such as roads, helps firefighters connect firelines to these locations.



THINNING FOR FOREST HEALTH



Fuel Reduction Methods

There are various ways to reduce or treat surface, ladder, and crown fuels to create fire-resistant forests, which are typically used in combination. For example, a stand may be thinned and pruned, and the resulting surface fuels piled and burned.

Thinning

Common questions about thinning include:

- Which trees should be selected?
- How far apart should trees be spaced?

Making decisions about thinning will involve a variety of other considerations.

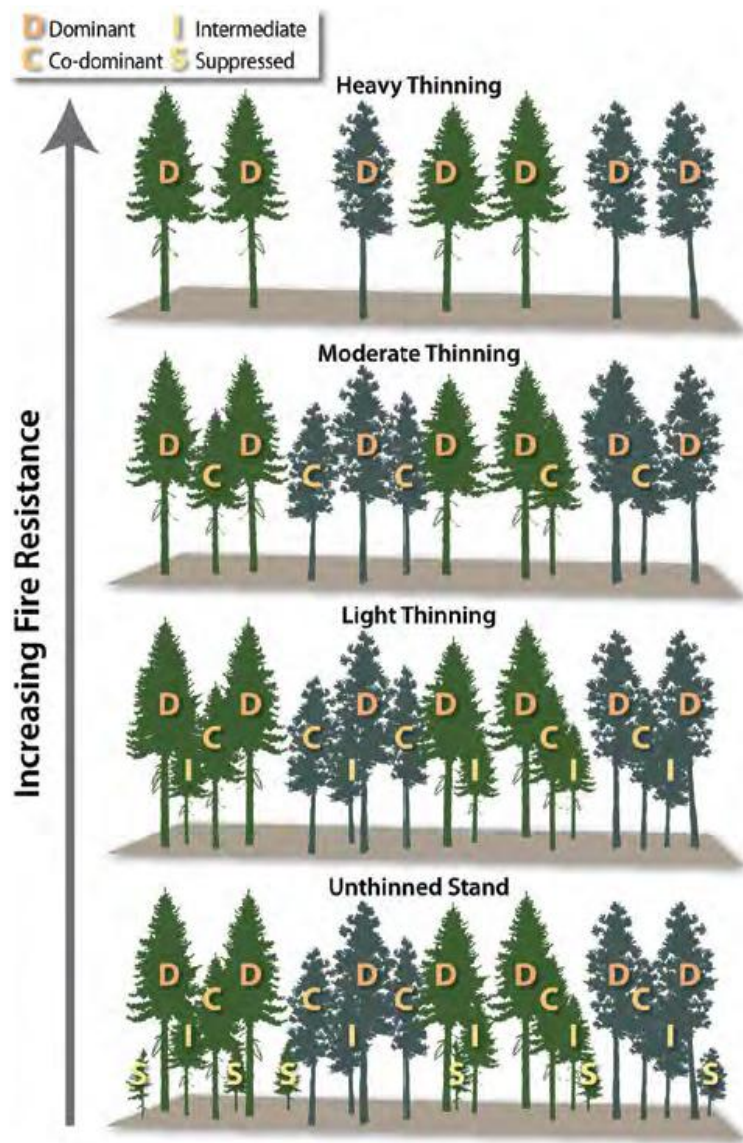


Figure 10

Tree selection

Remove smaller trees and retain larger, more vigorous trees (Figure 10). “Thinning-from-below,” removes ladder fuels, raises the base of tree crowns, and, if enough larger trees are removed, increases the spacing between tree crowns. Thinning from below is a common approach in even-aged stands.

When you want to maintain or promote a unevenly aged forest, a modified approach can be used. Trees can be thinned across the range of diameter or age classes so that stand density and ladder fuels are reduced while maintaining a multi-aged character.

Compared to an even-aged stand, such a stand will have a higher risk of crown fire because some younger understory trees (ladder fuels) would remain.

Pruning

Pruning can be combined with thinning or done as a stand-alone treatment. Pruning removes lower tree limbs, increasing the height of tree crown bases. A good height to shoot for from a fire-resistance standpoint is 3 to 4 m, though pruning even higher is beneficial.

The pruning slash should be disposed of through piling and burning, chipping, or, if fuel loads are light, cut and scatter.

To maintain tree health, pruning should leave at least a 50% live crown ratio (ratio of the length of the tree crown to the total height of the tree) and should not damage the tree bole.

Pruning is particularly effective in young stands where tree crowns have not yet lifted (the gradual death and branch shedding of lower tree branches from shading) on their own.



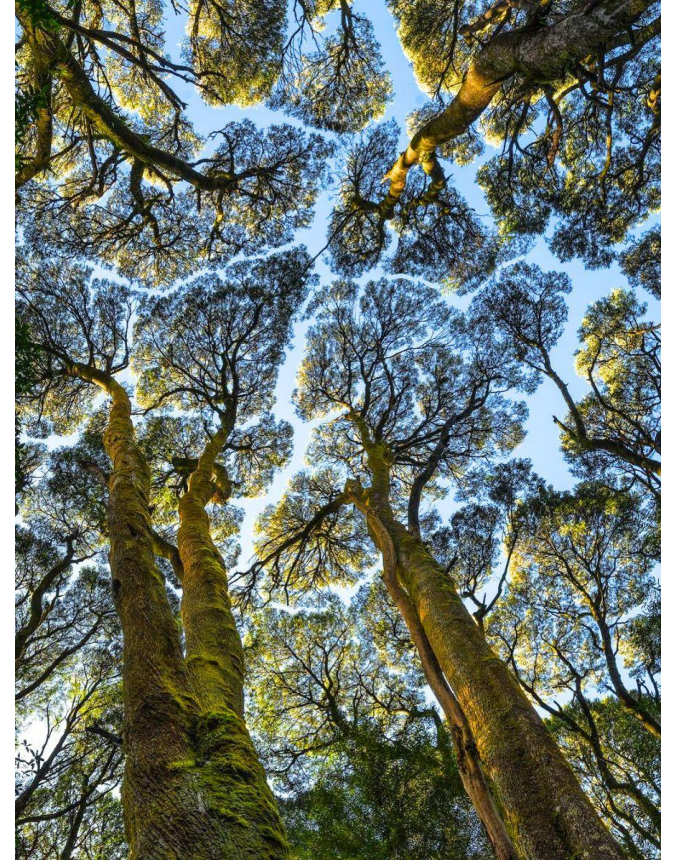
Tree spacing

If the branches of adjacent trees are overlapping within the stand, crown density is high enough to sustain crown fire under the right weather conditions. If trees are widely spaced, say with crowns spaced more than one dominant tree crown width apart, crown fires are much less likely to occur.

Factors that tend to increase the required crown spacing include steep slopes, locations with high winds, and the presence of species with dense, compact foliage.

Opening up the stand will dry surface fuels due to increased light levels, surface winds and temperatures. This may increase surface fire intensity and rate of spread unless total surface fuel loading is reduced.

Thinning that allows more light to reach the forest floor may result in the re-growth of small trees and shrubs, which over time become new ladder fuels.



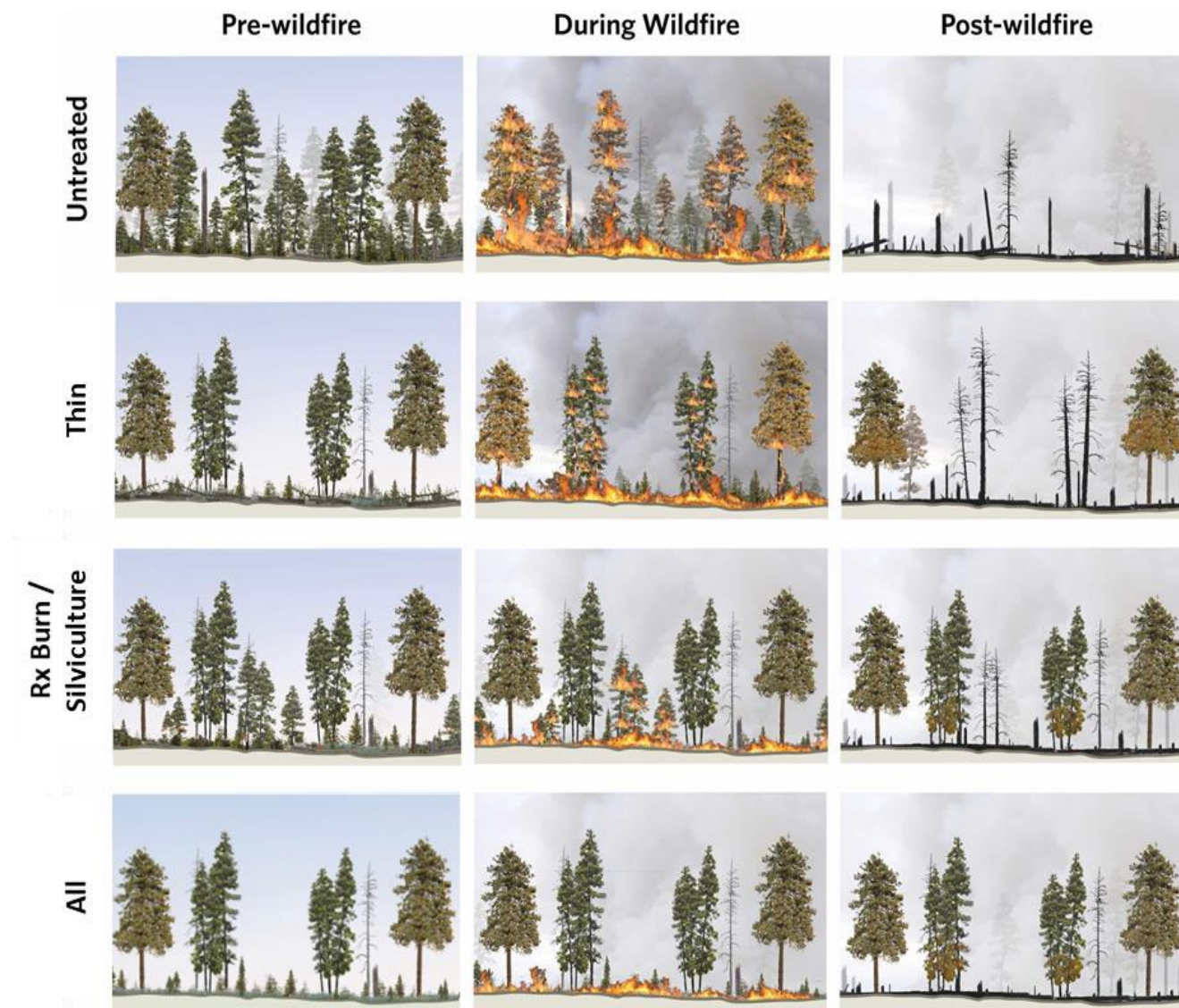


Fig. 1. Diagram demonstrating potential treatment effects on forest stand structure (pre-wildfire), fire behavior (during wildfire), and fire severity (post-wildfire).

Untreated stands are often **denser** than treated stands with **more ladder fuels** which leads to **higher fire intensity**, **increased risk of passive and active crown fire**, and **higher fire severity**.

“**Thin only**” stands have a **lower tree density**, but **high surface fuel loads** from slash left on site can in some cases lead to a **higher likelihood of passive crown fire**, **more intense surface fire**, and **higher fire severity** than other treatment types.

“**Rx burn**” treatments often retain **higher tree densities** than **thinned stands**, but due to **reductions in surface fuels** can still **reduce subsequent fire intensity and severity**.

“**Thin and prescribed burn**” treatments **reduce tree density**, **ladder fuels**, and **surface fuels** which leads to a **lower likelihood of passive and active crown fire**, **lower fire intensity**, and **lower fire severity**.

“**Thin and pile burn**” would be similar to “**thin and prescribed burn**”, but may retain higher surface fuels than areas treated with broadcast prescribed burns. Figure by Erica Sloniker.

NOTE: “Rx burn” = prescribed burn



Fig. 7. “Thin only” treatments are less effective at reducing wildfire severity than treatments that also utilize prescribed burning, with 68% of thin only treatments experiencing high severity in Oregon’s 2021 Bootleg Fire.

This photo shows a mixed conifer forest where the Klamath Tribes, The Nature Conservancy, and Lomakatsi Restoration Project worked with the US Forest Service to thin trees and apply prescribed burning as part of the climate adaptive ‘Black Hills Project’ where, in some areas, prescribed burning had not been applied yet.

Table 2. **Principles and appropriate sequencing** of thinning a forest stand to increase resilience to fire, i.e., “thinning from below”.

Principle	Effect	Advantages	Concerns
1. Reduce surface fuels	Reduces potential flame length	Facilitates fire control; less torching; reduced crown fire initiation/spread	Less surface disturbance with prescribed fire than other techniques
2. Increase height to live canopy	Requires longer flame length to initiate torching	Less canopy scorch and consumption	Opens understory; may allow surface wind to increase
3. Decrease crown density	Makes tree-to-tree crown fire less probable	Reduces crown fire potential	Surface wind may increase and surface fuels may be drier
4. Keep bigger trees of fire-resistant species	Less mortality for same fire intensity	Generally restores historic structure	Less economical; may keep trees at risk of insect attack

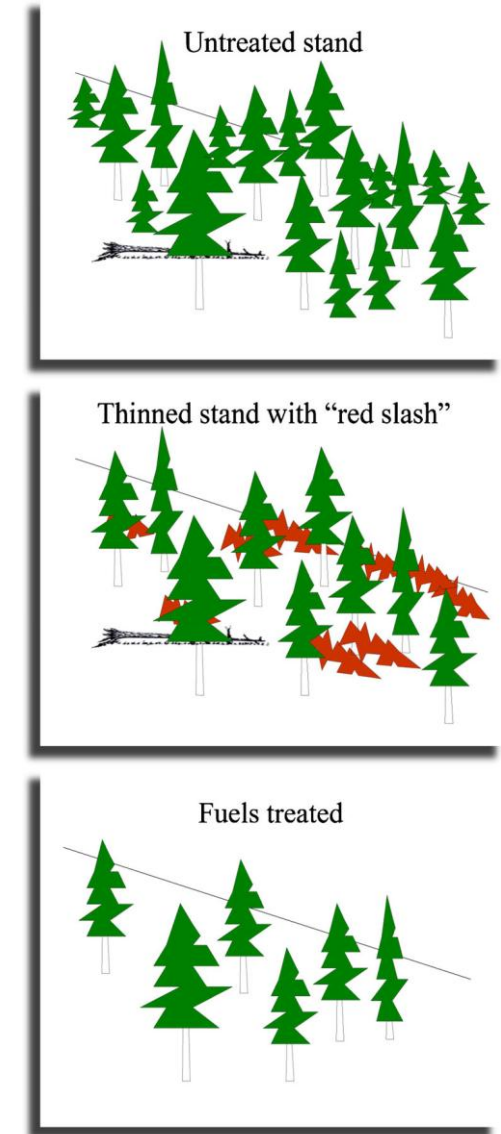


Fig. 2. A typical fuels treatment sequence involves reducing stand density by cutting small trees (“thinning from below”), followed by disposal of thinning slash, also known as “red slash” for the dessicated needles on cut trees. Fuels in the treated stand (bottom panel) can be maintained subsequently at desired levels with prescribed fire.

Table 1. Comparison of stand and landscape scale attributes of pyrosilviculture

Attribute	Stand Scale	Landscape Scale
Definition	Use fire directly to meet management objectives; alter silvicultural treatments to incorporate future prescribed fire.	Coordinate and consolidate treatments (mechanical, prescribed burn, managed wildfire) to reduce fuels and restore fire's ecosystem function.
Objectives	Create conditions that make prescribed fire feasible; sustain fuel conditions to encourage low- to moderate-severity wildfire.	Restore fire as an ecosystem process; minimize type conversion; leverage low/moderate severity fire areas as initial "treatments."
Operational means	Adjust planting and thinning to allow future fire use; prescribed burns at stand scale (<125 ac); scheduling around specific management goals.	Identify managed wildfire zones; implement anchor, ecosystem asset, and revenue treatments; broaden objectives to include heterogeneity and selection.
Measures	Fuel-load monitoring; wildfire behavior modeling; fire effects that enhance objectives (e.g., minimize crown damage).	General objectives derived from natural range of variability (NRV): tree density, structure, spatial pattern, fire behavior (% high-severity fire, patch size).
Limitations	Risk, resource, regulatory barriers; outcome variability vs. non-fire treatments; perception of fire conflicting with timber goals.	Crew and equipment availability; increased smoke days; liability; institutional caution.
Opportunities	Use leaf area index, relative density index; small burns during winter droughts or cool nights; multiple stand types allow flexible fire application windows.	Treat large areas where prescribed fire, managed wildfire, and mechanical treatments are synergistic; provide habitat for sensitive species.

Landscape-level management

Main Principles of Fuel Management at the Landscape Scale

1. Reduce Fuel Loads

- **Goal:** Decrease the amount of combustible material available to burn.
- **Methods:** Mechanical thinning, prescribed burning, grazing.

2. Alter Fuel Structure

- **Goal:** Modify vertical and horizontal fuel continuity to reduce fire intensity and spread.
- **Methods:**
 - Vertical separation (ladder fuel removal) to reduce crown fire risk.
 - Horizontal discontinuity to compartmentalize the landscape.



Main Principles of Fuel Management at the Landscape Scale

3. Create Fuelbreaks and Defensible Spaces

- **Goal:** Slow fire spread and **facilitate suppression operations.**
- **Design:** Strategic strips of reduced or modified fuels (natural or artificial), often located near roads, ridgelines, or built assets.

4. Strategic Location of Treatments

- **Goal:** Maximize landscape-level effectiveness with limited resources.
- **Approach:** Use fire behavior modeling and historical fire data to identify high-risk areas (e.g., near communities or fire-prone corridors).

5. Landscape Mosaics and Pyrodiversity

- **Goal:** Increase spatial heterogeneity in fuels and vegetation types.
- **Benefit:** Reduces the potential for large, high-intensity fires and supports biodiversity.



Main Principles of Fuel Management at the Landscape Scale

6. Integration with Land Use and Conservation Goals

- **Balance** fire prevention with ecological and social values (biodiversity, C storage, recreation).
- **Collaborative planning** across jurisdictions and landownerships is essential.

7. Adaptive Management and Monitoring

- **Track effectiveness** of fuel treatments over time.
- **Adjust** strategies based on outcomes, new data, and climate/fire regime changes.



Main types of landscape-scale fuel treatment

- Two main types:
 - Linear – fuelbreaks (strategic / primary) and perimetral / secondary)
 - Areal – patch mosaics
- **Strategic fuelbreaks:** broad, placed on ridgelines, cover large areas (100s to 1000s ha) and aim at containing the spread of large wildfires. Occupy fixed positions in the landscape and are designed to have an effect beyond the fuel treatment area.
- **Perimetral fuelbreaks:** narrower, located around urban areas and infrastructure. Designed to protect localized, valuable assets. Intensely managed.
- **Patch mosaics:** small to medium-sized areas of varied shapes. The fuel treatment effect is local, reducing fire rate of spread, intensity, and severity. The mosaic is dynamic, generating a patchwork of fuel ages that constrains fire behavior.

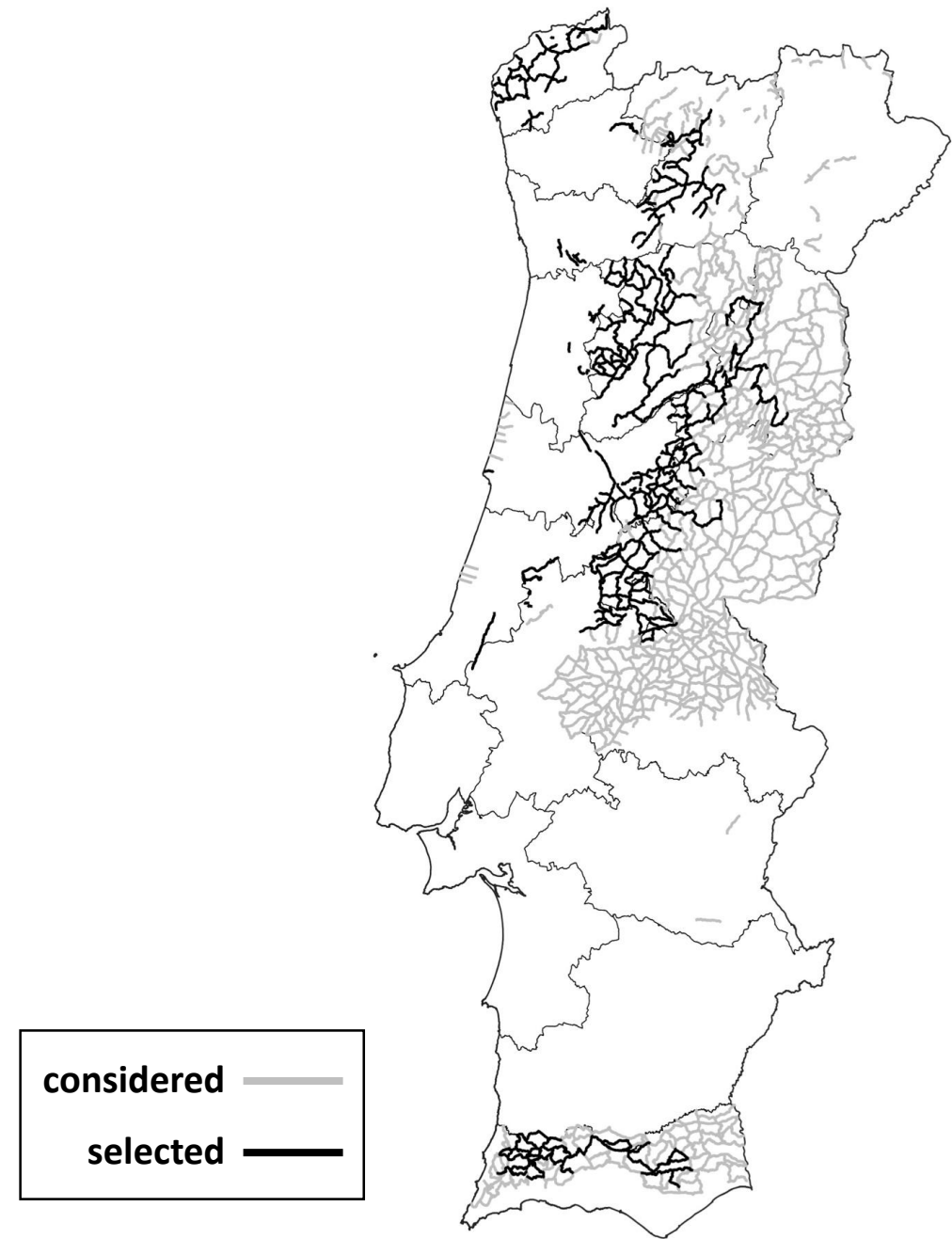
Linear vs. area-based landscape-scale fuel treatments

Feature	Linear Fuel Management (Fuelbreaks)	Area / Patch-Based Treatments
Form	Narrow, elongated strips (e.g., along roads or ridgelines)	Larger, discrete blocks or patches spread across the landscape
Primary Objective	Fire spread containment, access for suppression	Reduce fire intensity, severity, and rate of spread across the treated area
Mechanism	Interrupt horizontal fuel continuity and facilitate firefighting	Modify fuel loads and structure to influence fire behavior locally
Design Focus	Strategic placement (e.g., perimeter of high-value areas, choke points)	Spatial arrangement for mosaic effects and fuel heterogeneity
Width / Size	Typically tens of meters wide	Can range from a few to hundreds of hectares
Implementation	Often easier to maintain; may integrate with roads/trails	More intensive in terms of treatment effort and planning
Suppression Support	Designed to support active fire suppression (safe zones, access)	May indirectly support suppression by reducing flame length and spotting potential

Primary fuelbreak network

Primary Fuelbreak Network

- The Primary Fuelbreak Network (PFN) is designed to reduce the area affected by large wildfires.
- It facilitates direct fire suppression by creating favorable conditions for firefighting operations at strategically selected locations.
- In Portugal, primary fuelbreaks typically have a standard width of 126 meters and divide the landscape into compartments ranging from approximately 500 to 10,000 hectares.
- The planned PFN for Portugal will extend around 3,000 kilometers, covering a total area of approximately 38,000 hectares.
- It is under construction and is estimated to have a total cost of 120 M€.



Primary Fuelbreak Network

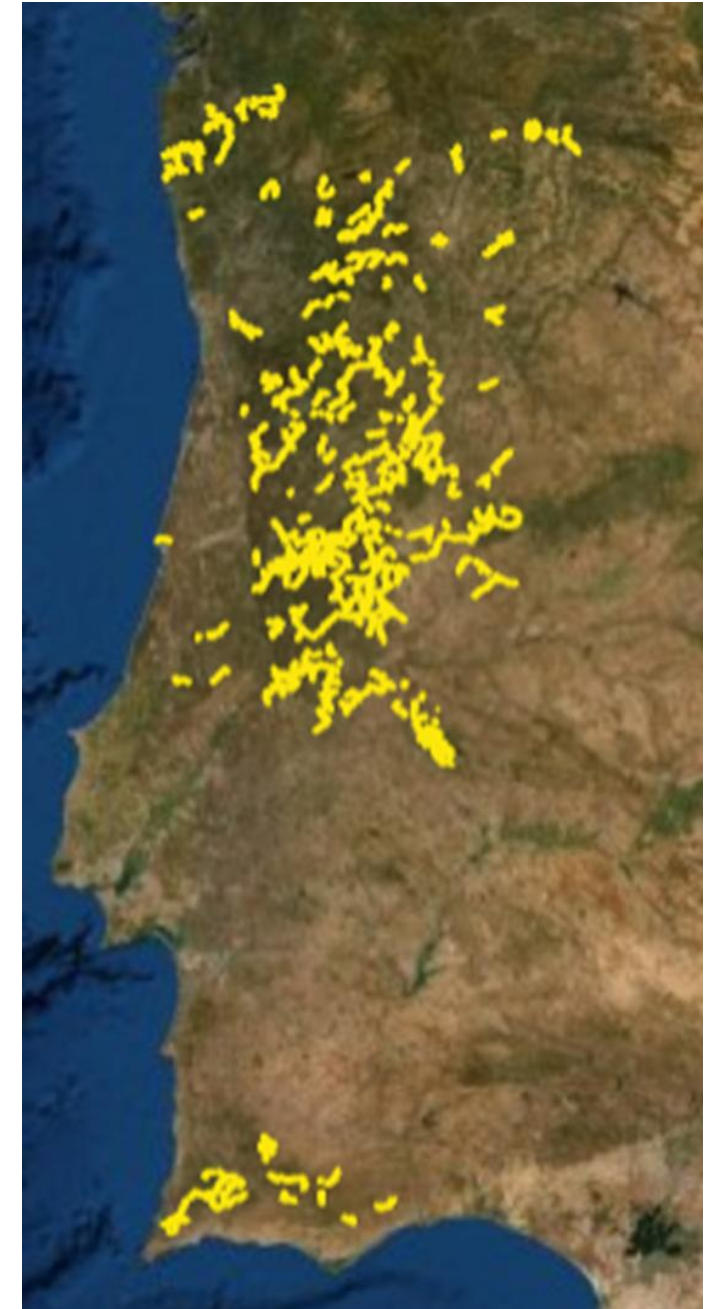
- Placed at fixed positions, often along ridges.
- These strips aim to compartmentalize the landscape and provide protection to areas beyond those directly treated.
- Fuelbreaks do not need to be totally cleared of vegetation and may even have tree cover, however at very low density (“shaded fuelbreaks”). They may be economically productive under pastoral use, beekeeping, etc.
- Fuelbreaks need periodic maintenance to remain effective, at the rate of about every 5 years in humid Mediterranean regions.



Primary Fuelbreak Network

Planning, implementation, and maintenance of the primary fuelbreak network must take into account:

- effectiveness at suppressing large wildfires,
- safety of firefighting personnel,
- socioeconomic, landscape, and ecological value of rural areas,
- terrain features of the local landscape,
- history of large wildfires in the region and their expected behavior under severe fire weather conditions,
- economic activities within the fuelbreaks, contributing to their technical and financial sustainability.



Viana do Castelo



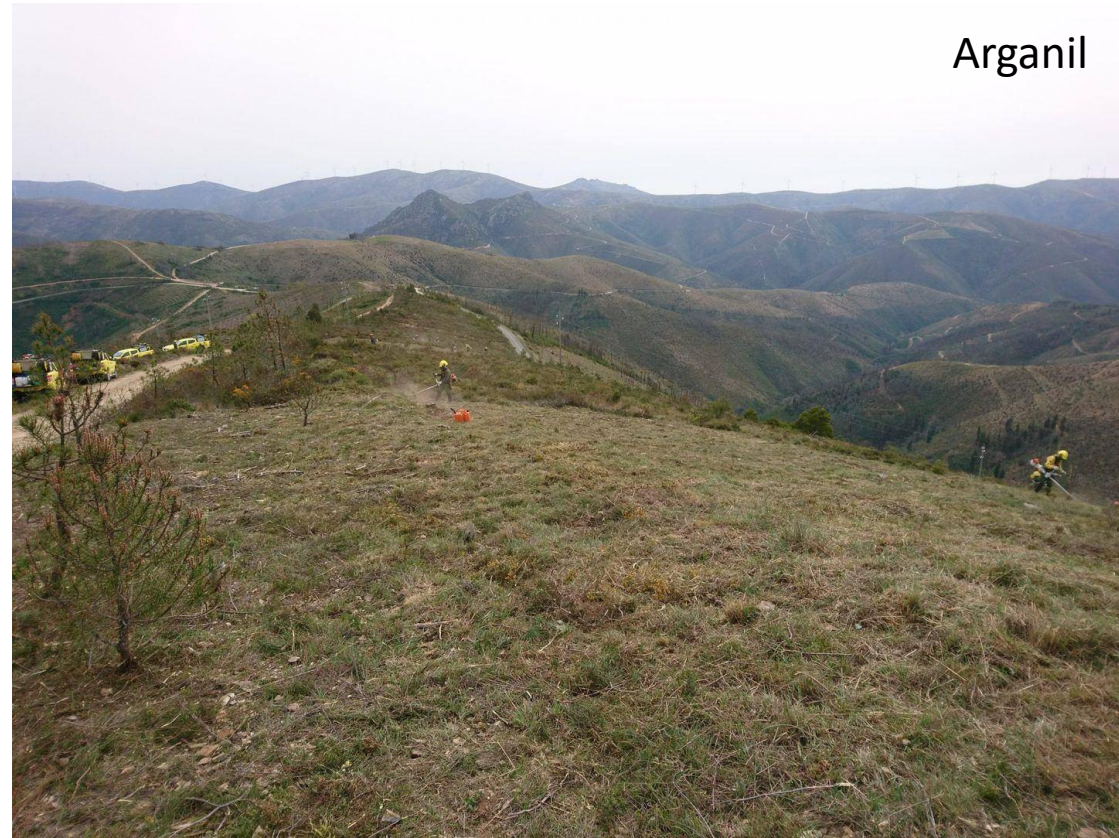
Loulé



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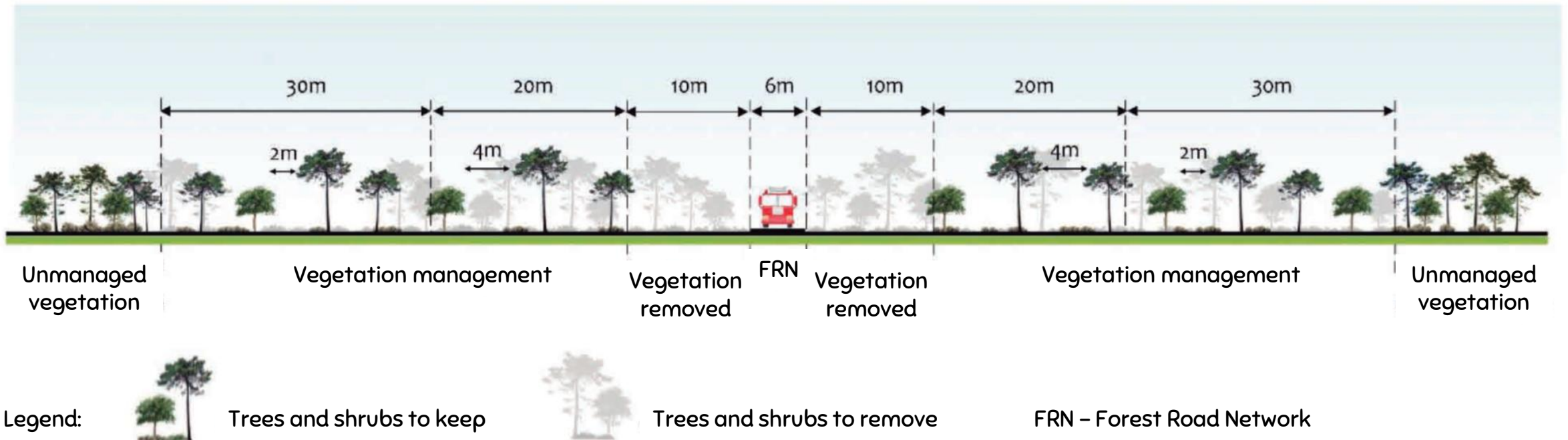
Silves

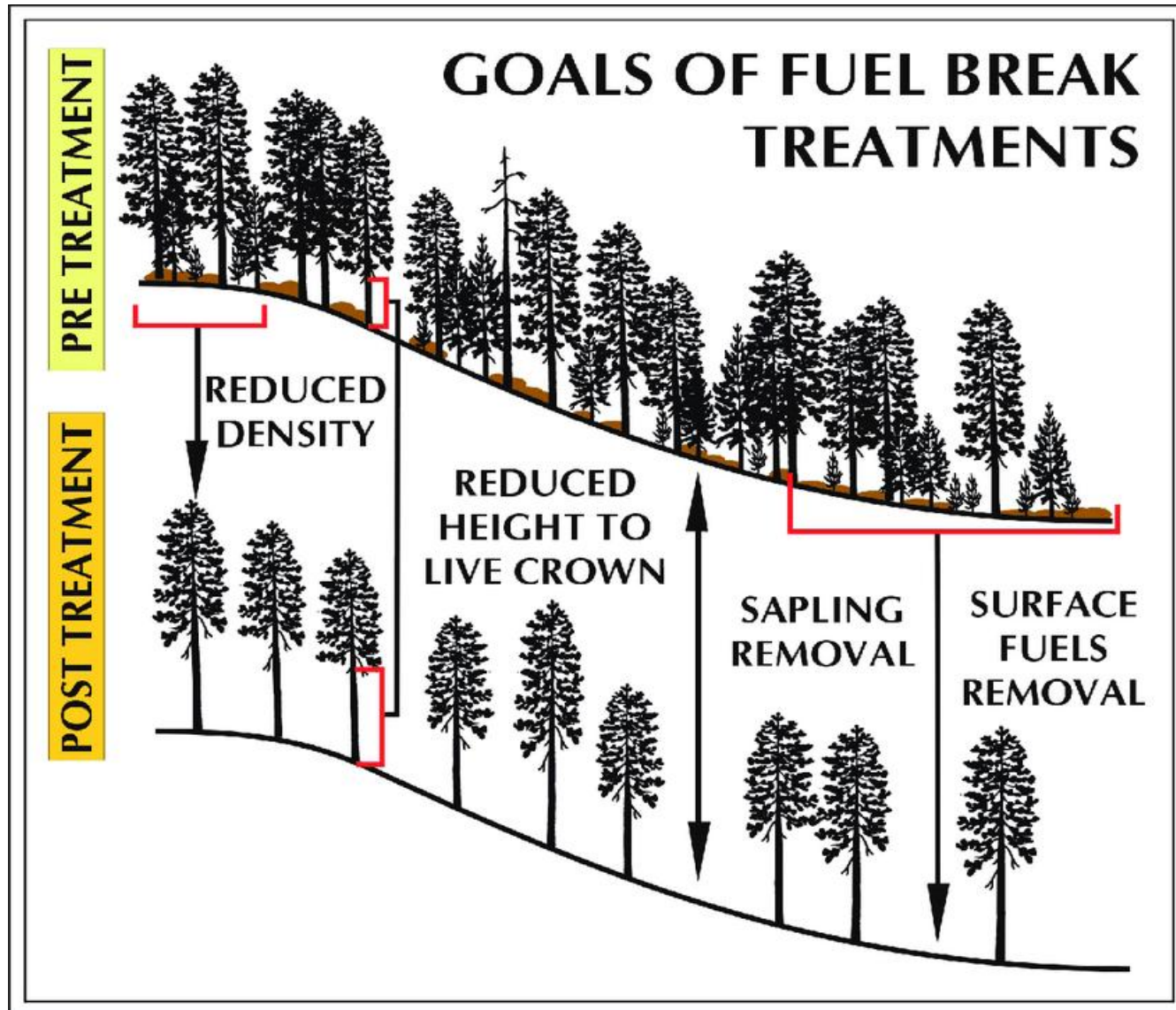


Cross-sectional layout of a **standard Primary Fuelbreak**. It admits the presence of sparse tree cover outside of a 20 m buffer on each side of the road, for economic and aesthetic reasons, i.e. it is a **shaded fuelbreak**.

Road access is mandatory at all PFN segments to ensure safe and effective fire suppression interventions.

PFN segments may be wider in specific terrain setting, such as steep slopes and mountain passes (saddles).

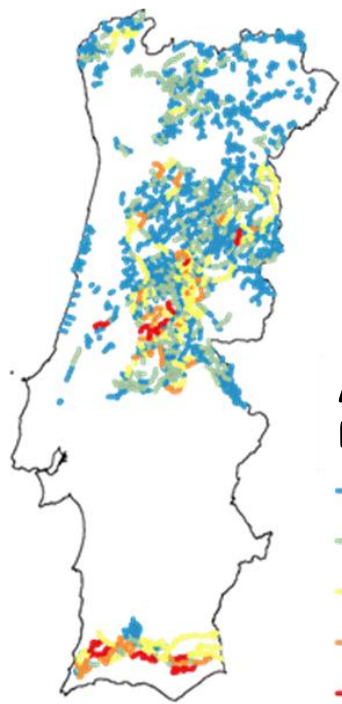




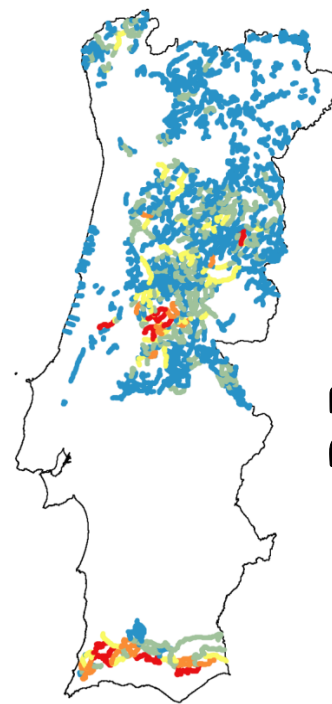
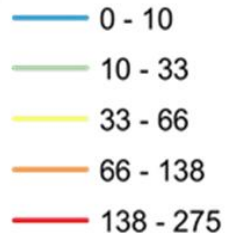
Shaded fuelbreaks typically result from thinning a forest stand located on a ridge.

Their use is uncommon in Portugal, which is regrettable because properly designed shaded fuel breaks have economic and aesthetic advantages.

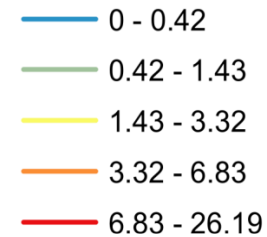
They should be the preferred design in areas with nature protection status.



Avoided area burned per FBN segment (ha/yr)



Residential buildings protected per FBN segment (#/yr)



High priority segments for each goal



Simulation-based analysis of PFBN performance for two major priorities of the National Plan for Integrated Rural Fire Management developed at CEF.

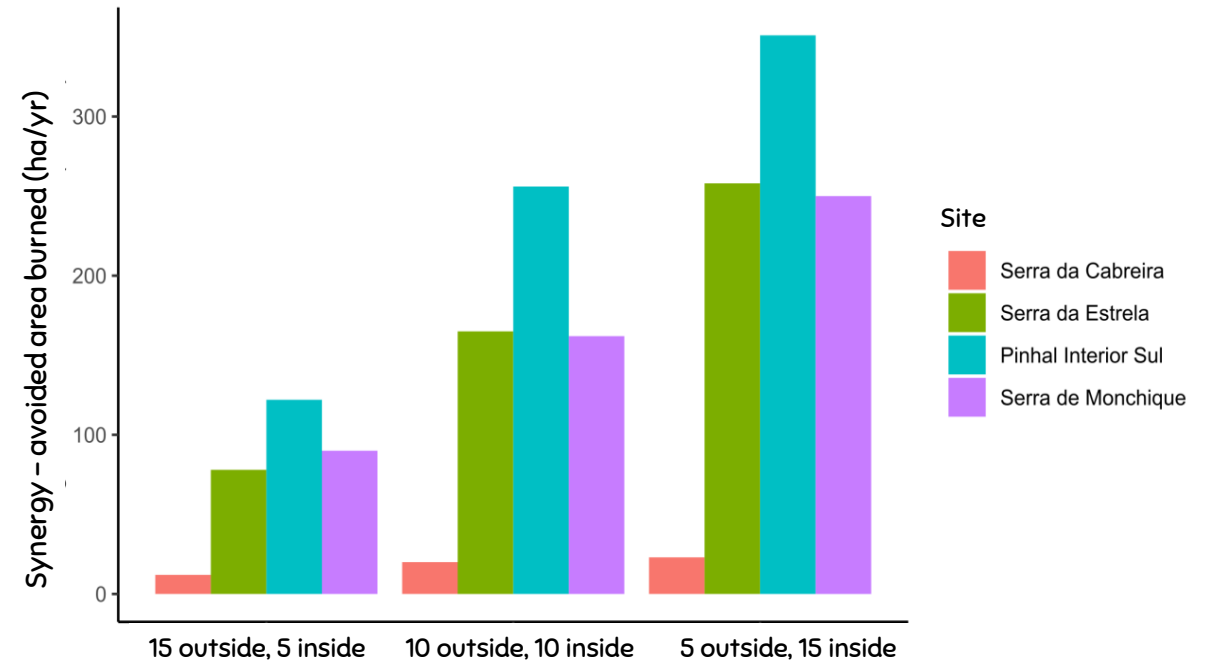
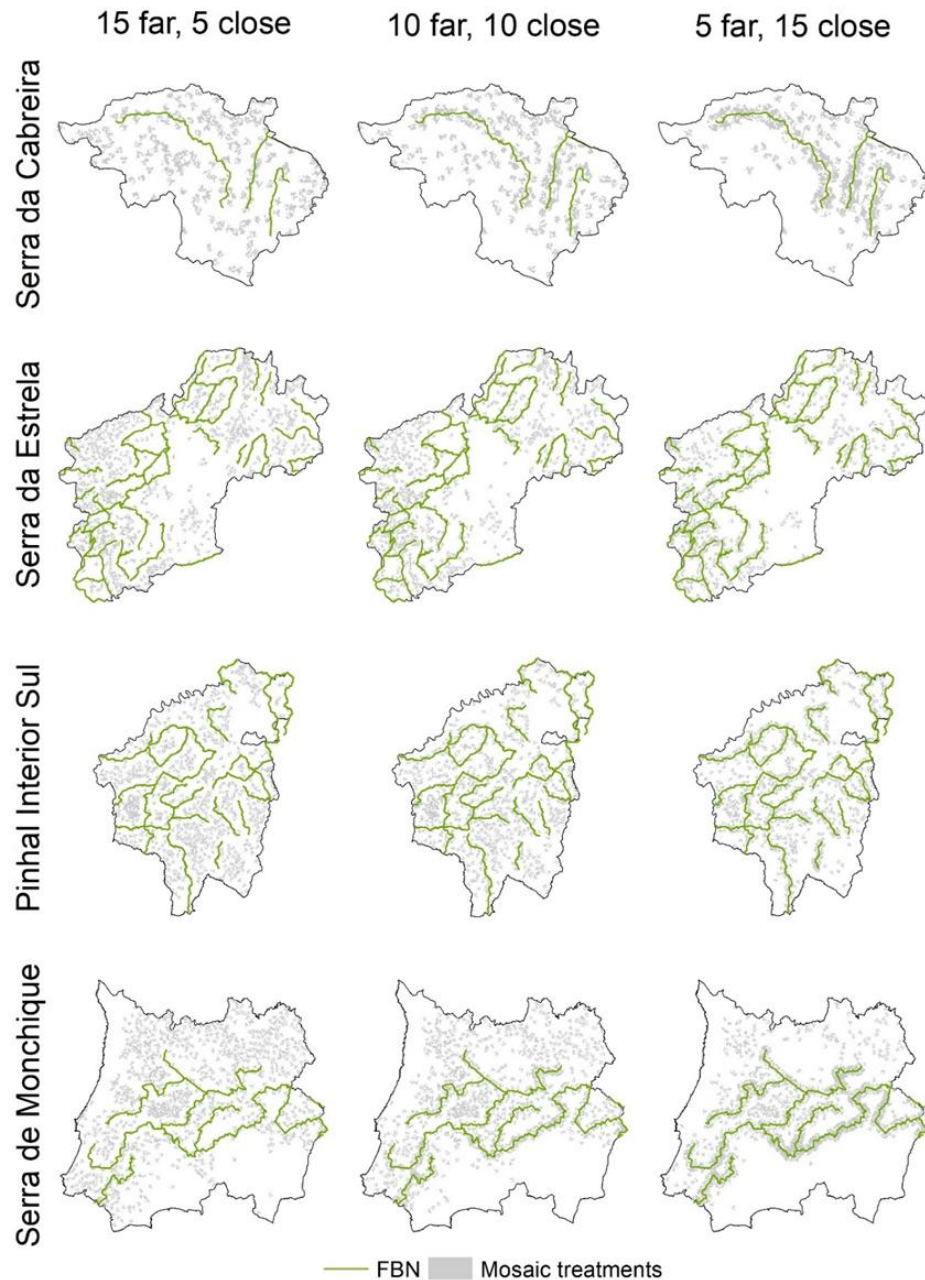


Figure 40 – Synergistic effect, expressed as the annual reduction in burned area (in hectares), for each mosaic management scenario and pilot area. The synergistic effect is defined as the difference between the combined reduction in burned area achieved when two strategies are implemented jointly and the sum of the reductions achieved when each strategy is applied individually.

Locating fuel treatment patches close to the primary fuelbreak network (PFBN) slows down the fires and decreases their intensity, increasing the PFBN effectiveness.

Firefighters fail to use the Primary Fuelbreak Network. They stay close to urban areas and on roads, which causes them to miss opportunities to suppress fires.

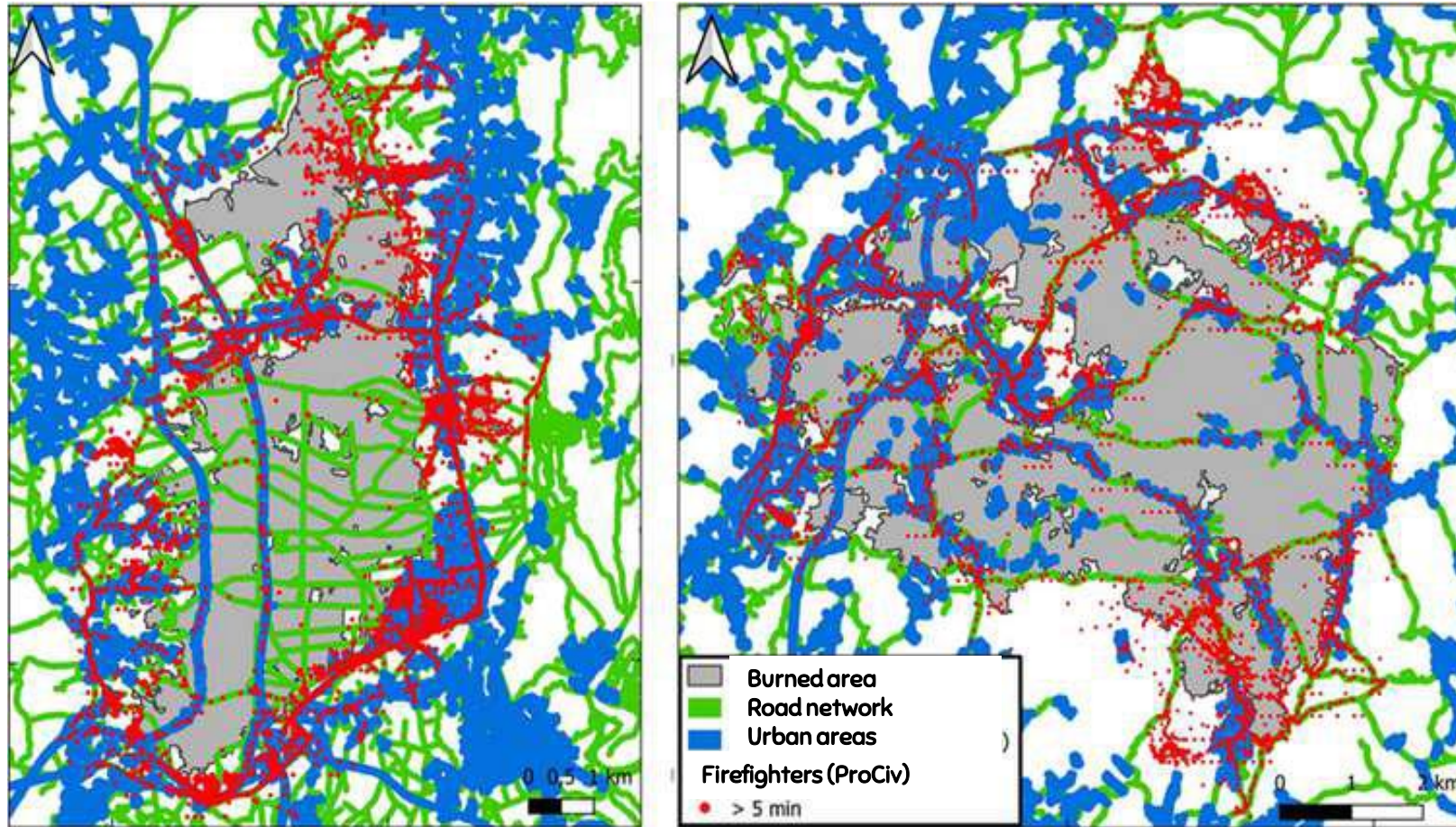


Figure A15. Spatial distribution of personnel and vehicle SIRESP data from Volunteer Firefighter Corps and Civil Protection Special Force, over the full duration of the Albergaria-a-Velha (left) and Leiria (right) 2022 wildfires.

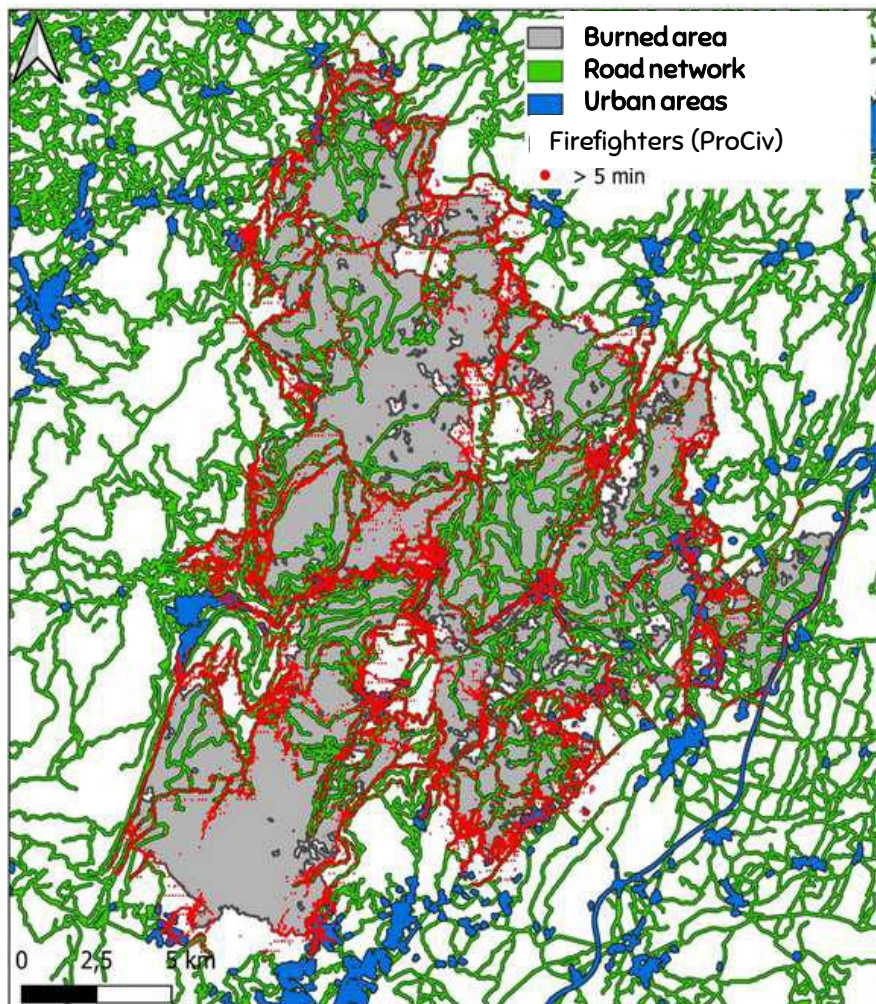


Figure A14. Spatial distribution of personnel and vehicle SIRESP data from Volunteer Firefighter Corps and Civil Protection Special Force, over the full duration of the Serra da Estrela 2022 wildfire, August 6 – 20.

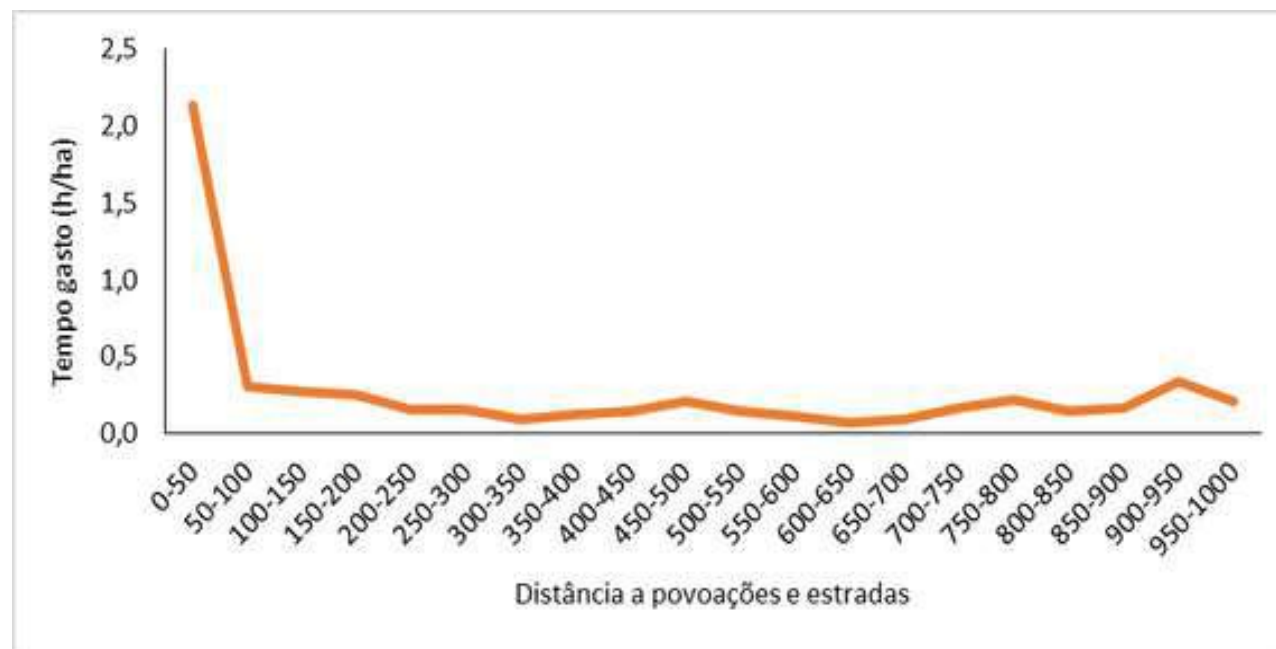


Figure A16. Time spent per hectare by Volunteer Firefighter Corps and Civil Protection Special Force, as a function of distance to urban areas and roads. Vehicle data and stationary signals > 2 hours were excluded. Serra da Estrela 2022 fire.

RUI and the secondary
fuelbreak network

Wildland-Urban interface is the area where human-built environments meet and mix with natural vegetation, and where the potential for wildfire impacts on people, property, and ecosystems is heightened (FAO definition).



In predominantly humanized landscapes the concept is often designated Rural-Urban interface (RUI). Delineation of the RUI is useful to assist the establishment and management of the Secondary Fuelbreak Network (SFN).

Monchique 2018



Vieira de Leiria 2017



Pedrógão Grande 2017



Alvares 2017



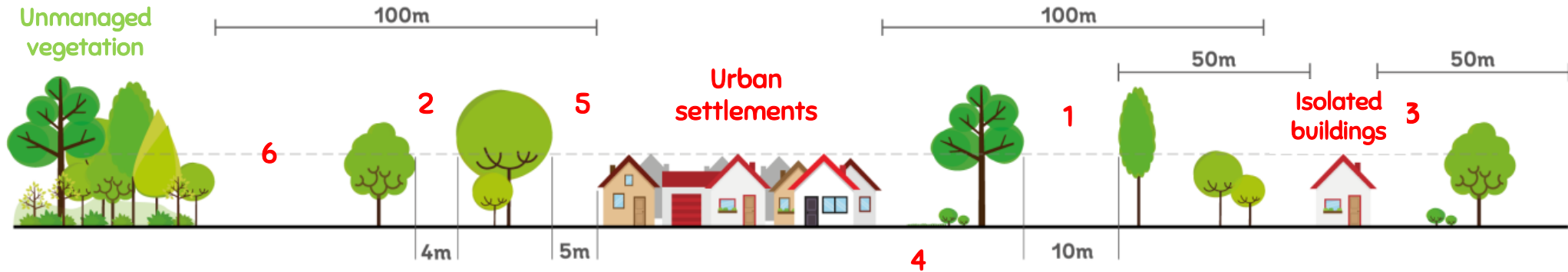
Fires in the RUI

The Secondary Fuelbreak Network is a **perimetral network of cleared or managed vegetation strips** designed to reduce wildfire intensity and slow fire spread, especially **near areas at risk** (like towns, critical infrastructure, or important natural areas).

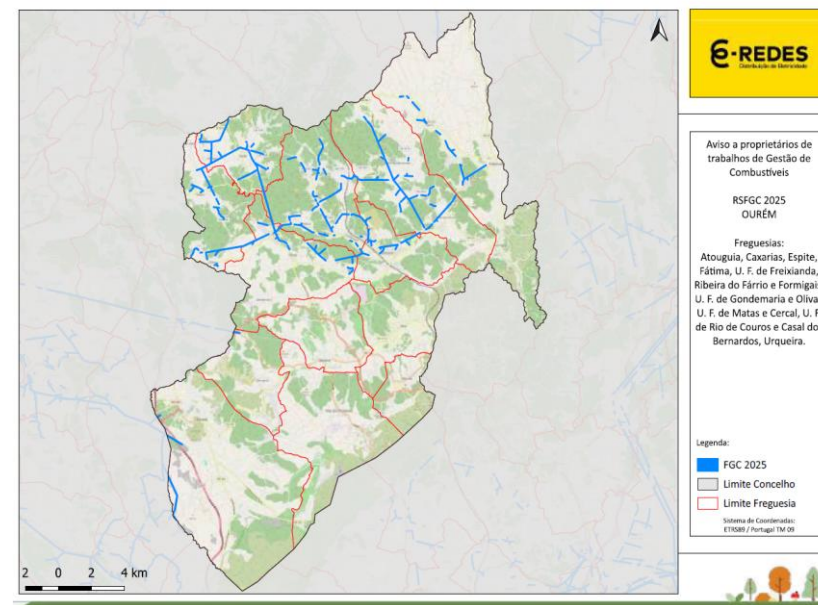
- The network components are **strategically located** – not randomly – based on wildfire risk maps, landscape features, and protection priorities.
- They **reduce available vegetation** (fuel) by activities like thinning, clearing, pruning, or controlled burning.
- They help **protect people, houses, roads, powerlines**, and other infrastructure.
- "Secondary" means they **complement the Primary Network**, which usually includes broader, regional-scale fire breaks, mostly located along ridgelines.
- Their creation and maintenance is **legally mandated in Portugal** under national wildfire prevention plans, like the **National Plan for Integrated Management of Rural Fires** (*Plano Nacional de Gestão Integrada de Fogos Rurais*).

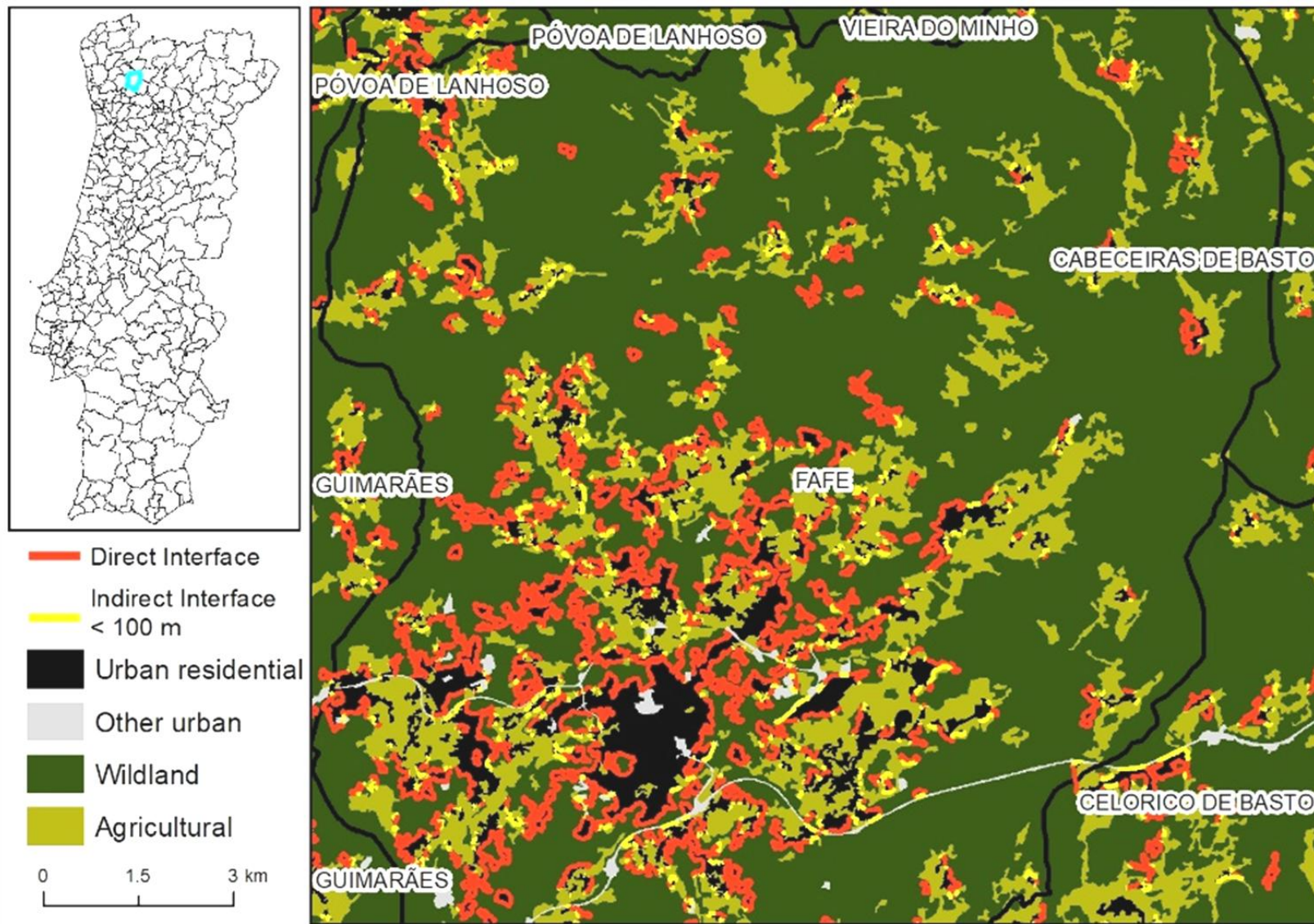


Cross-sectional layout of a Secondary Fuelbreak Network segment



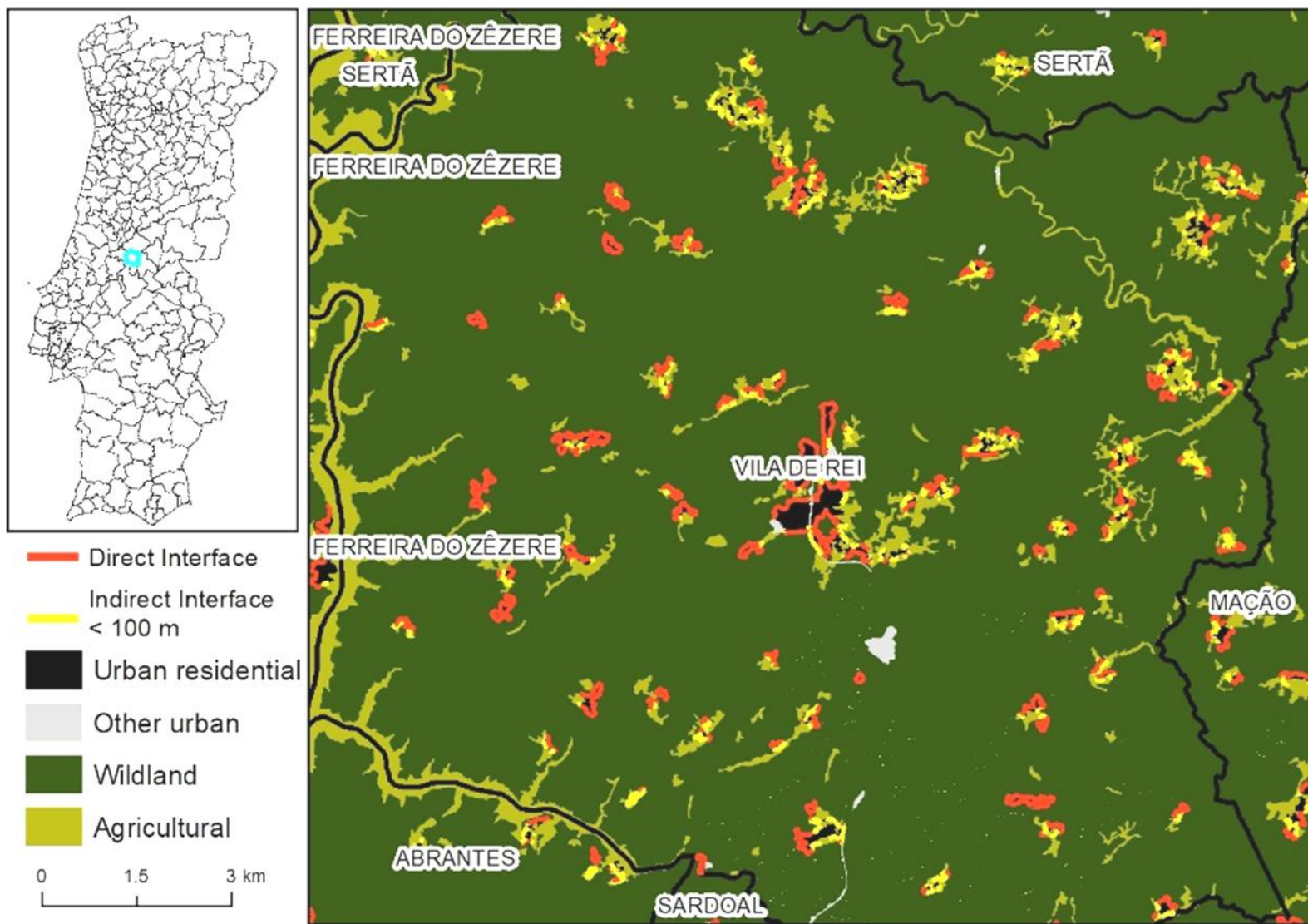
1. In stands of maritime pine and eucalyptus, the minimum distance between tree canopies must be 10 meters.
2. For other species, the minimum canopy distance must be 4 meters.
3. Trees must be pruned to half their height until they reach 8 meters in height, after which pruning must reach 4 meters above the ground.
4. In the shrub layer, the maximum vegetation height must not exceed 50 cm, and in the sub-shrub layer, it must not exceed 20 cm.
5. The tree and shrub canopies must be at least 5 meters away from buildings, and they must not overhang building roofs.
6. No accumulations of combustible materials – such as firewood, timber, forest or agricultural residues, or other flammable substances – are allowed.

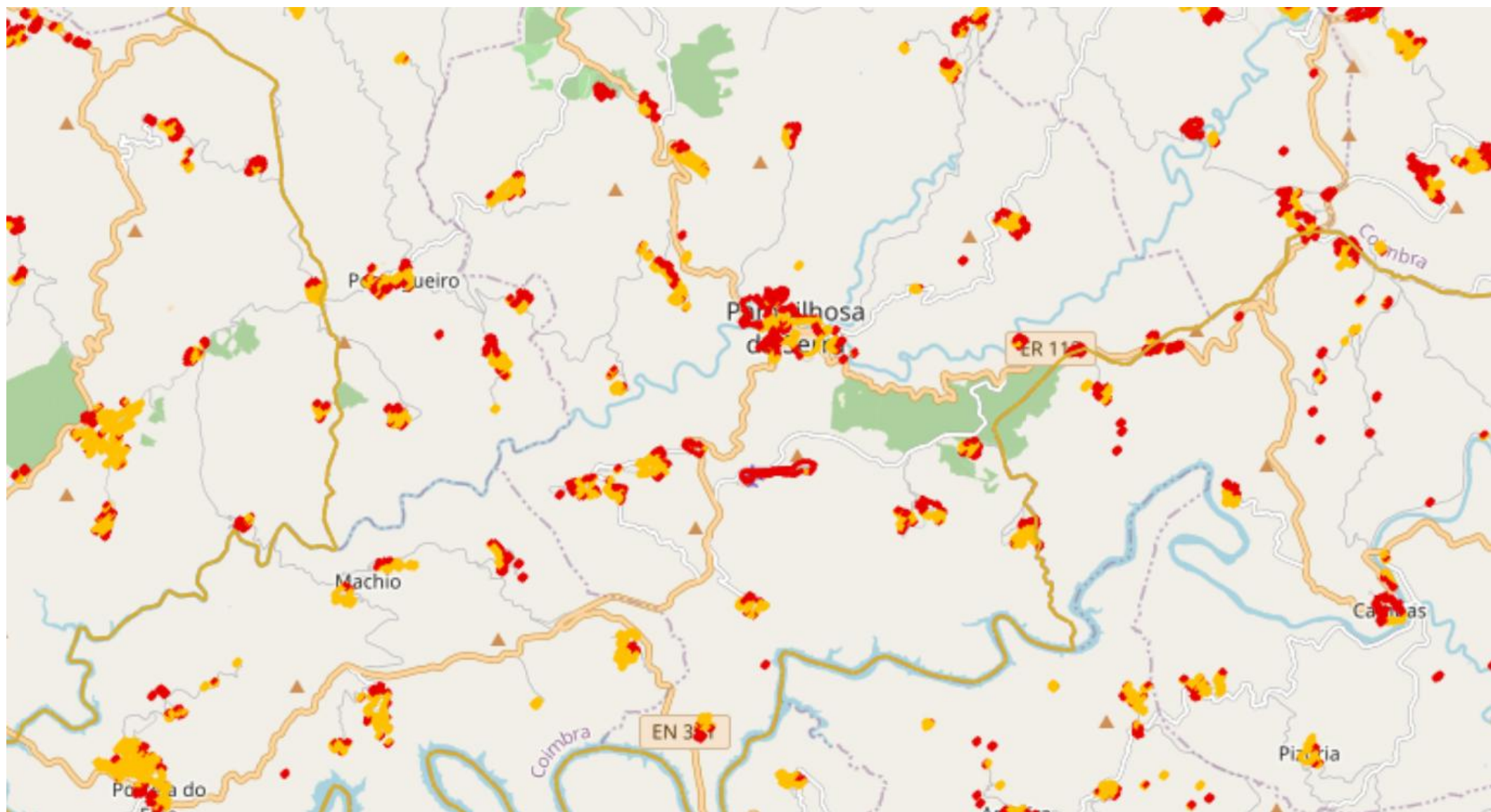


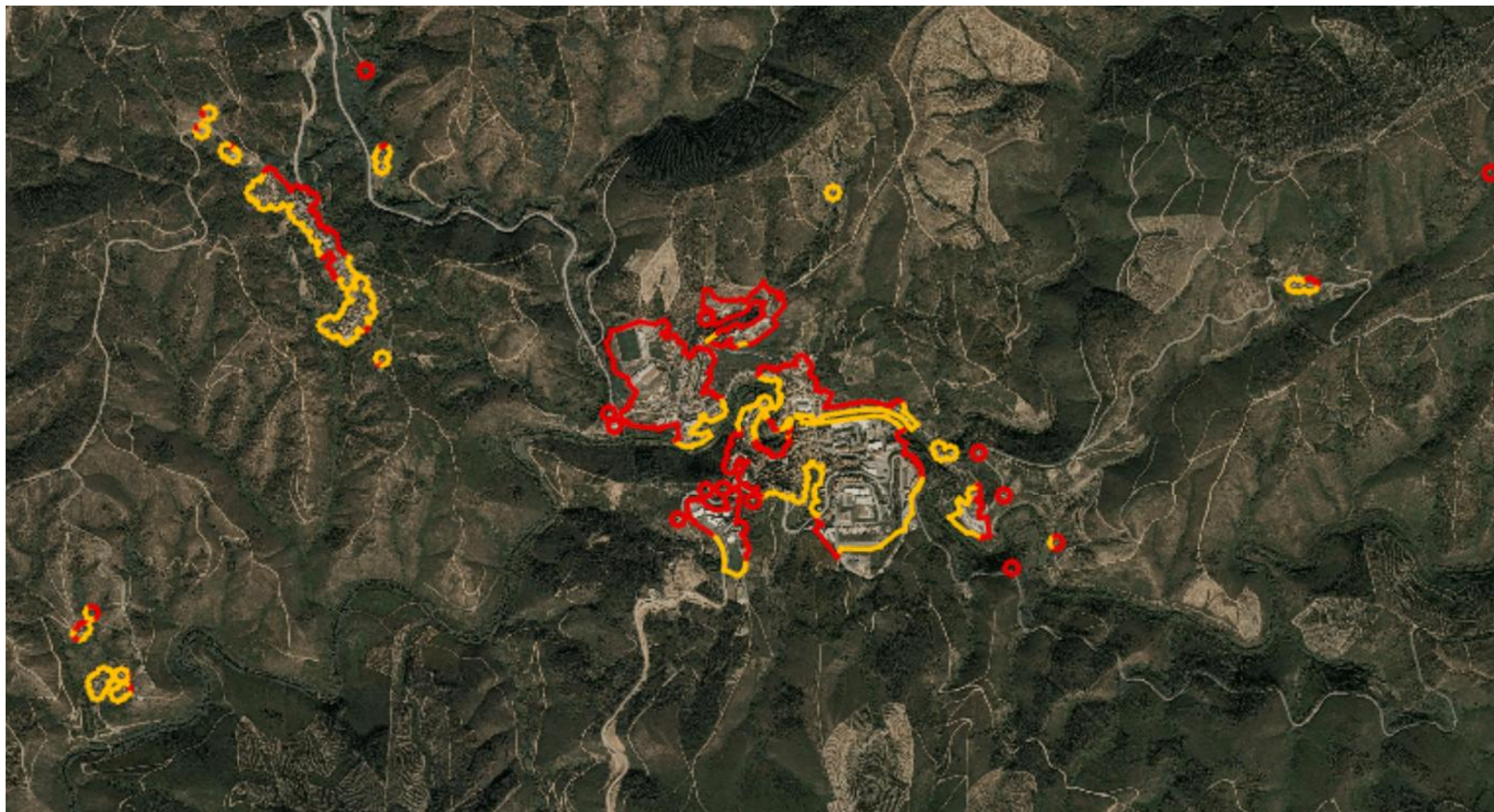


Direct interface: fire-prone vegetation adjacent to urban area.

Indirect interface: fire-prone vegetation < 100 m away from urban area.







Wildland-urban interface in Portugal (2018)

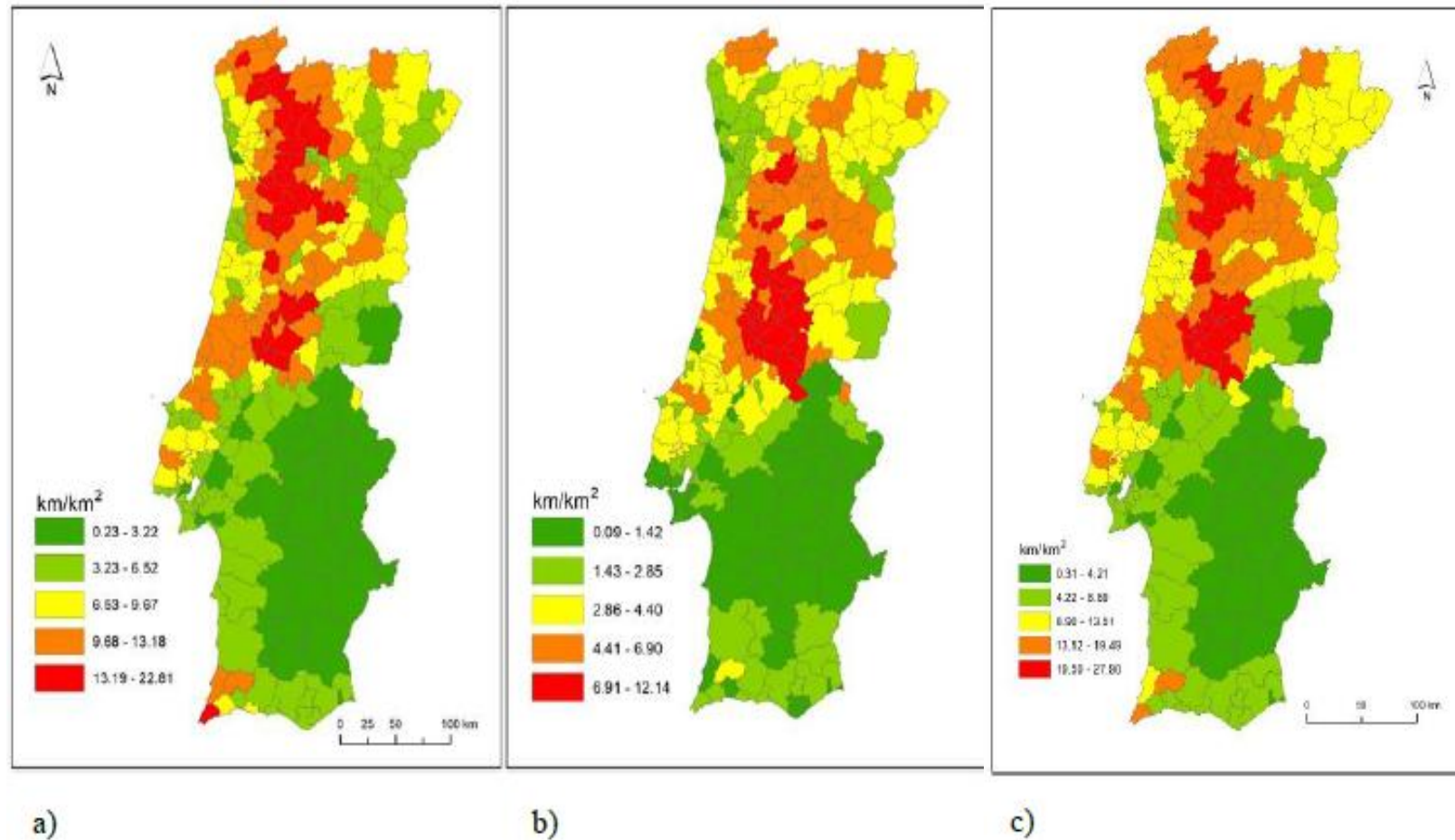


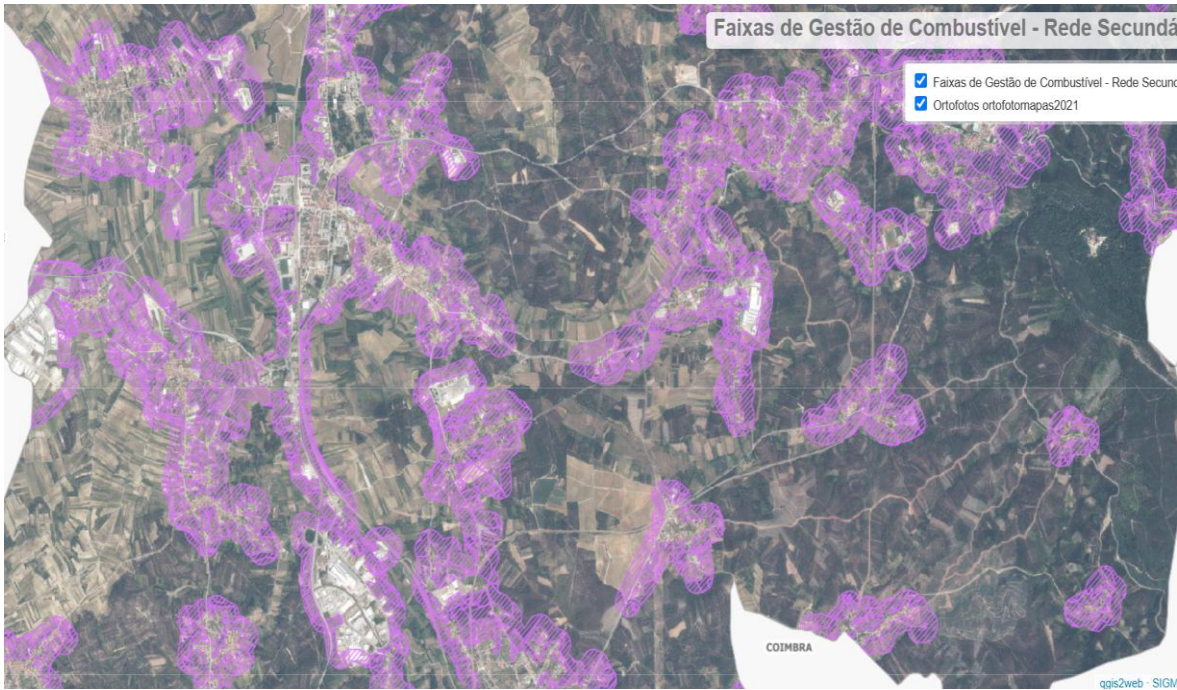
Figure 5. Direct (a), indirect (b) and total (direct + indirect) WUI density.

Secondary Fuelbreak Network

The secondary fuel management strip network serves to reduce the impact of wildfire spread, passively protecting transport routes, infrastructure and public facilities, built-up areas, and forest and agricultural formations of special value. It also functions to isolate potential ignition sources.

The secondary network is established around the following infrastructures:

- Road network (10 meters);
- Railway network (10 meters);
- Power transmission lines (3-10 meters);
- Gas and petroleum pipeline networks (≥ 7 meters);
- Infrastructure supporting the emergency telecommunications system (7 meters);
- Campsites, hotels, business parks, industrial establishments, fuel stations, logistics platforms, energy or gas production and storage facilities, and landfills (100 meters);
- Built-up areas adjacent to forest land (100 meters);
- Residential, commercial, and industrial buildings located less than 50 meters from forest land (50 meters) and from agricultural land (10 meters).



Secondary Fuelbreak Network at Mealhada municipality, Aveiro district.

Design of the SFN can be improved if based on an efficient delineation of the WUI.

A novel definition of the WUI is the set of line segments where urban areas are in immediate physical contact with fire-prone vegetation (direct interface), or are less than a pre-specified distance away from fire-prone vegetation (indirect interface).

The linear WUI is particularly appropriate for the highly humanized landscapes of Europe, where transitions between patches with distinct land uses typically are sharp, often with simple geometric forms.

Secondary (perimeter) fuelbreaks

- Secondary fuelbreaks (SFB) are placed around or near homes, settlements, and other assets at the Wildland–Urban Interface.
- Their purpose is to protect life and property by reducing fire intensity and ember load near structures.
- SFB are narrower, and more intensively managed than primary fuelbreaks; they may include irrigation, landscaping, and reliance on fire-resistant vegetation.

Secondary (perimeter) fuelbreaks

- Secondary fuelbreaks are created around infrastructure networks for roads, railways, power transmission lines, gas and petroleum pipelines, and emergency telecommunications lines.
- Wider secondary fuelbreaks protect campsites, hotels, business parks, industrial establishments, fuel stations, logistics platforms, energy or gas production and storage facilities, landfills, and built-up areas adjacent to forest land.
- Residential, commercial, and industrial buildings located less than 50 meters from forest land (50 meters) and from agricultural land (10 meters).

Patch mosaics

Fuel treatment patches:

- Mosaics of low fuel loading patches that reduce fire rate of spread, intensity, and severity over the treated areas.
- They are particularly appropriate for shrubland and replicate ancestral practices of traditional pastoralists, who often use fire to renovate pastures.
- When specifically designed for fire prevention they may be strategically placed to hinder fire spread over historically fire-prone areas.



Fuel treatment patches:

- Burn patches are not supposed to have fixed positions, the mosaic changes in space and time every year.
- It is a mosaic of fuel age classes with varying degrees of flammability. Each location in a treatment area ought to be burned ca. every 5 years.
- In Portugal, “baldios” – commonly owned lands in mountainous regions – are privileged territories for this type of landscape-scale fuel management.
- The Portuguese Forest Service (ICNF) has a program (MARQ) to work with shepherds in the design and execution of prescribed burns mosaics to renovate pastures and reduce fire hazard.



Cairngorms, Scotland 2018



Northern Territory,
Australia 2024



Prescribed burns in the tropical savanna of the Northern Territory, Australia.

Burned patches are arranged in irregular, ring-like structures, suggestive of linear fuelbreaks.

The pattern is a sort of hybrid between linear fuelbreaks and patch burns.

In this predominantly flat land, the ring-like arrangement provides the landscape compartmentalization that the ridges of watersheds naturally support in mountainous terrain.

In the following fire season, details of the burn pattern sizes, shapes, and locations will change.



Early dry season burns, Northern Territory, Australia. June 2020.

Complementarity between fuelbreaks and patches at the Landscape Scale

Fuelbreaks and patches **work best together** when strategically integrated:

- **Fuelbreaks slow or halt fire spread along key corridors**, enabling fire fighters to intervene.
- **Patch treatments alter fire behavior across the broader landscape**, reducing the likelihood that fires will reach or breach fuelbreaks with high intensity.



Complementarity between fuelbreaks and patches at the Landscape Scale

Strategic Coupling

- Fuelbreaks can be **buffered** by nearby patch treatments to increase their effectiveness.
- Patch treatments can be located to **steer fires** toward fuelbreaks or low-risk zones.

Ecological & Risk Considerations

- Patch-based treatments can promote **ecological heterogeneity** (pyrodiversity), while fuelbreaks risk being ecological barriers unless properly designed.
- Combining both enables **risk reduction** without homogenizing the landscape or compromising biodiversity.



Policies, plans, and
responses

Table 2a. Summary of main findings on landscape–wildfire interactions in Mediterranean Europe, challenges to address and proposed policy and landscape management responses.

Main findings	Challenges to address	Policy / Landscape management responses
Land cover changes are increasing fire hazard	Reversing the detrimental implications of land cover changes	Implement policies to reverse population decline in mountain areas
		Promote agricultural and pastoral activities, particularly in mountain regions
	Managing land use and land cover to reduce fire hazard	Promote farming practices in areas surrounding villages, and enforce existing laws on fuel management around buildings and villages
		Stop afforestation policies in high fire hazard areas
		Reduce fuel accumulation through prescribed/pastoral burning
		Explore potential for biofuels as a tool to remove fuels from the landscape
		Forest planning should give preference to deciduous broadleaved and mixed forests, rather than pine plantations

Table 2b. Summary of main findings on landscape–wildfire interactions in Mediterranean Europe, challenges to address and proposed policy and landscape management responses.

Main findings	Challenges to address	Policy / Landscape management responses
Large fires are becoming more frequent	Being prepared for large fires and aim at reduced damage, rather than area burned	Invest more in prevention (fuel management) rather than suppression policies
		Focus on reducing fire intensity and potential damage, rather than trying to control fire size, specially under extreme fire weather conditions
		Reduce vulnerability of values at risk (landscape planning at the rural/urban interface, fuel management around buildings and villages)
		Educate citizens on fire–response planning (defence and evacuation)

Table 2c. Summary of main findings on landscape–wildfire interactions in Mediterranean Europe, challenges to address and proposed policy and landscape management responses.

Main findings	Challenges to address	Policy / Landscape management responses
Frequent fires are creating homogeneous landscapes covered with fire-prone shrublands	Increase landscape heterogeneity and decrease the probability of repeated burning	In landscape planning take into account that specific topographic locations (e.g. steeper slopes, ridge crests, south exposures) may have greater fire hazard
		Promote landscape heterogeneity, by planning land covers with alternate higher and low fire hazard
		In forested areas, plan forest management to spatially alternate stands with different ages/fuel types
		Implement policies for the wise use of fire in pastoral burnings
		Increase fire prevention measures in forests that have no post-fire regeneration ability (e.g. in immature <i>Pinus halepensis</i> and <i>P. pinaster</i> forests, and in vulnerable <i>P. nigra</i> and <i>P. sylvestris</i> forests)
		Increase fire prevention in growing oak forests, which have long recover times

Table 2d. Summary of main findings on landscape–wildfire interactions in Mediterranean Europe, challenges to address and proposed policy and landscape management responses.

Main findings	Challenges to address	Policy / Landscape management responses
Fuel management may be successful only if fire weather conditions are not extreme	Combine landscape compartmentalization with self-protection options	Under normal fire weather conditions, promote fuel management to extend fire suppression options and effectiveness
		Under extreme fire weather, focus on reducing fire intensity and damage through self-protection
		Plan fuel isolation options at the landscape level, combined with area-wide fuel modifications
		Use the whole range of fuel management techniques, including silviculture, agricultural and pastoral management, and the use of fire as a tool

PLANO NACIONAL DE GESTÃO INTEGRADA DE FOGOS RURAIS 20-30

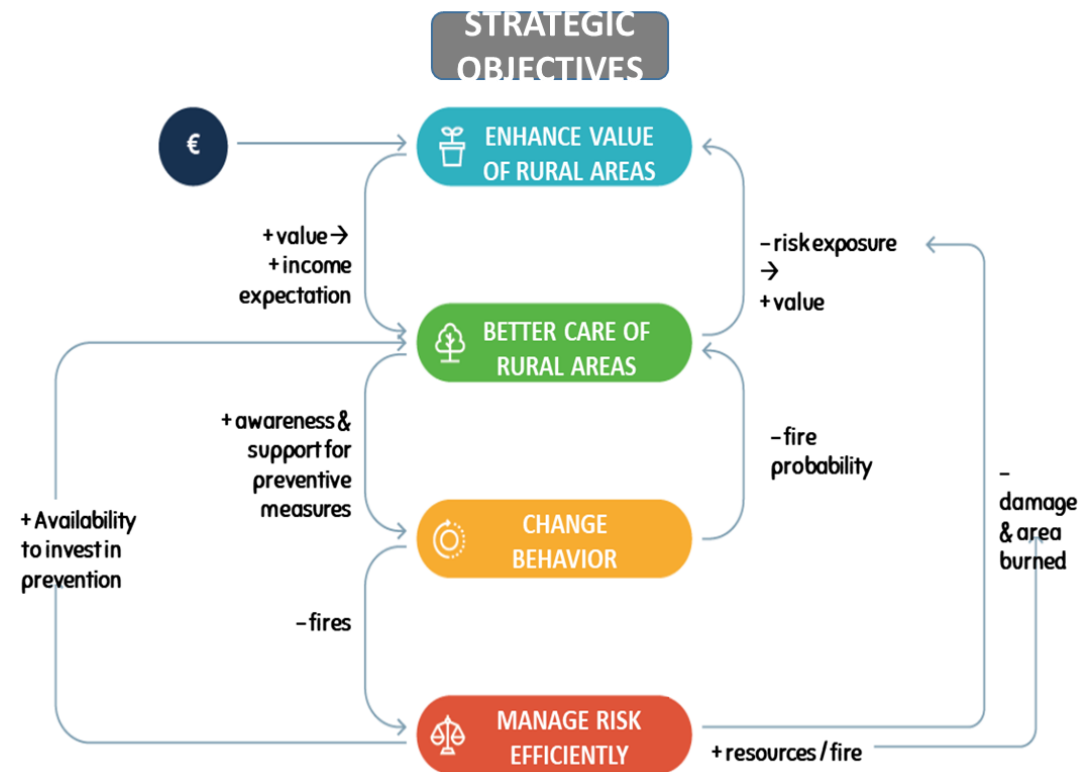
National Plan for Integrated
Management of Rural Fires

NOVEMBRO 2022



VISION: Protect Portugal from severe wildfires

MISSION: Protect people and assets from rural wildfires, support rural development, and promote ecosystem conservation.



PLANO NACIONAL DE GESTÃO INTEGRADA DE FOGOS RURAIS 20-30

PROGRAMA NACIONAL DE AÇÃO

04 Junho 2021



National Action Programme

- 4 Strategic Orientations
- 11 Strategic Objectives
- 28 Programmes
- 97 Projects

The 4 Strategic Orientations

- Enhance Value of Rural Areas
- Care for Rural Areas
- Change Behaviors
- Manage Risk Efficiently

